慢性腎臟病與透析患者之抗結核藥物

劑量調整與臨床藥事照護

115年6月臨床藥學會雲嘉區雙月會 | 6月25日

張綜瑋 藥師

彰化基督教醫療財團法人 員林基督教醫院 藥劑課

嘉南藥理大學 藥學系 **兼任講師**

課程綱要

結核藥物的特性

- 抗肺結核藥物
- 潛伏結核藥物

特殊族群在結核藥的劑量調整

- 腎功能
- 肝功能
- 體重
- HIV藥物交互作用
- COVID藥物交互作用
- 移植藥物交互作用

INH+RMP+PZA+EMB

2個月

INH+RMP+PZA

4個月

INH+RMP

7個月

如果無法使用Rifamycin➔視為多重抗藥的治療，治療期間會比較長

結核藥物治療藥物特性

空腹
發燒跟倦怠
容易交叉抗藥
87%
LTBI
腸胃不適

	Isoniazid	Rifampin	Rifabutin	Rifapentine	Pyrazinamide	Ethambutol
機轉	抑制細胞壁合成	抑制RNA polymerase	抑制RNA polymerase (較少交互作用)	抑制RNA polymerase (長效)	活化成 Pyrazinoic acid	抑制細胞壁合成
劑量	5 mg/kg/day	10mg/kg/day	300 mg/day	1200 mg/day 600 mg (一週1-2次)	25 mg/kg/day	20 mg/kg/day
懷孕 第二孕期開始治療	**安全使用**	**安全使用**	經驗有限 建議避免	資料有限 不建議常規使用	**WHO建議可用** **慎選用於重症(可)**	**安全使用**
哺乳 母乳中藥物濃度低	**可哺乳**	**可哺乳**	無資料	無資料	**可哺乳**	**可哺乳**
肝功能限制	慎用，可能須避免 (肝毒性高)	可用於Child A Child B、C應避免	資料有限	資料有限	肝毒性高 肝硬化者應避免	可用，較不會有肝毒性
腎功能限制	無須調整 糖尿病、腎功能不全、營養不良、慢性肝病、酗酒、感染人類免疫缺乏病毒（HIV）、末梢神經炎、懷孕、哺乳	無須調整	無須調整	資料有限	**eGFR <30 時應延長劑量間隔至每週3次** **(洗腎後給)**	**eGFR <30 時應延長劑量間隔至每週3次** **(洗腎後給)**
不良反應	肝毒性 週邊神經病變 **特殊族群給Pyridoxine**	肝毒性、血液異常 體液橘紅	**白血球低下(2%)** 葡萄膜炎(8%)交互作用	發燒、過敏反應 藥物交互作用	肝毒性、**高尿酸血症** 關節痛	**視神經炎(1-2%)** 視力模糊
交互作用	抗癲癇藥、抗黴菌藥 Phenytonin 濃度下降	**抗凝血藥物** HIV、避孕藥	較少誘導酵素，適合與部分抗HIV藥併用	HIV	尿酸<13可先觀察 少見，但需注意與尿酸藥物交互作用	少交互作用，需監測視力影響藥物合併使用

排斥藥物
有交互作用，但不要刻意避開

二線結核藥物治療藥物特性

	Streptomycin	Cycloserine	Prothionamide	Para-aminosalicylate	Ethionamide
機轉	抑制蛋白質合成	抑制細胞壁合成	抑制mycolic acid合成	抑制葉酸代謝	抑制mycolic acid合成
劑量	15 mg/kg/day	起始250 mg QD，逐漸增量至500 mg BID（max）	起始250 mg QD，逐漸增量至500 mg BID	起始2 g BID，逐步增量至4 g BID	起始250 mg QD，逐步增加至500 mg BID
懷孕	不建議使用 (耳毒性)	有CNS毒性，除非無更佳替代，不建議	存在肝毒性及CNS毒性 懷孕期應避免使用(致畸胎)	資料不足 懷孕期使用需審慎	有潛在神經毒性和肝毒性，非首選
哺乳	可分泌至母乳 建議避免使用	可進入乳汁 應避免或監測嬰兒中毒症狀	資料有限	資料有限	建議避免使用或監測嬰兒毒性症狀
肝功能限制	較安全（非肝代謝主途徑）	相對安全 但肝功能不全應減量 (治療抗藥性結核或副作用)	有肝毒性風險 肝病患者應避免或減量	有肝毒性 與ethionamide併用風險更高	有明顯肝毒性 肝硬化或肝炎不建議使用
腎功能限制	腎功能不全應減量	腎功能不全應減量	腎功能不全應減量	腎功能障礙者建議減量 多數資料建議避免使用	腎毒性較低 但資料有限，建議監測
不良反應	耳毒性、腎毒性	精神症狀、周邊神經病變，需合併pyridoxine	噁心、嘔吐、金屬味、肝毒性、甲狀腺功能低下、精神症狀	噁心、腹瀉、肝毒性 甲狀腺低下症	噁心、肝毒性、視神經病變 甲狀腺功能低下
交互作用	與其他耳毒/腎毒性藥物交互作用	與乙醇或CNS抑制藥交互作用，增加神經毒性	與PAS合併會增加甲狀腺功能低下風險	與ethionamide併用易導致甲狀腺功能低下及腸胃副作用	與PAS併用易致甲狀腺功能低下及肝毒性增加

表 6-1 第一線抗結核藥物-單方藥物

藥物 (縮寫)	毒性/副作用	給藥方式	每日建議劑量依體重 (mg/Kg)(容許範圍) Max (最大劑量)	一般用法劑量	
				50Kg 以下	50Kg 以上
isoniazid（INH）	肝、神經、皮膚敏感	口服或肌肉/靜脈注射	5（4-6）Max: 300 mg	100 mg/tab 2-3 顆	100 mg/tab 3 顆
rifampin（RMP）	體液/尿液變橘、肝、血液、胃腸不適、皮膚敏感	口服或靜脈注射	10（8-12）Max: 600 mg	150 mg/tab 3 顆	300 mg/tab 2 顆
rifabutin(RFB)	肝、白血球低下、皮膚敏感、眼葡萄膜炎(uveitis)	口服	5 Max:300mg	≦30 kg: 150 mg 1 顆 31~45 kg: 150 mg 1.5 顆 ≧46 kg: 150 mg 2 顆	
pyrazinamide（PZA）	肝、高尿酸血症	口服	25(15-30)Max: 2000 mg	40~55 kg: 500 mg/tab 2 顆 56~75 kg: 500 mg/tab 3 顆 ≧76 kg: 500 mg/tab 4 顆	
ethambutol（EMB）	視神經炎	口服	15（15-20）Max:1600 mg	40~55 kg: 400 mg/tab 2 顆 56~75 kg: 400 mg/tab 3 顆 ≧76 kg: 400 mg/tab 4 顆	
streptomycin（SM）*	耳毒性、腎毒性、暈眩或聽力障礙	肌肉/靜脈注射	15(15-20) Max:1000 mg	500~750 mg	750 mg ~1000 mg

* 累積總劑量不超過 120 gm

表 6-2 第一線抗結核藥物-複方藥物

藥物（縮寫）	毒性/副作用	給藥方式	一般用法劑量	
			50Kg 以下	50Kg 以上
Rifinah150 (RFN)(RMP 150 mg + INH 100 mg) Rifinah300/RINA/-Macox Plus (RMP 300 mg + INH 150 mg)	肝、血液、胃腸不適、 皮膚敏感	口服	【Rifinah150】3 錠	---
			---	【Rifinah300】2 錠
Rifater (RFT) (RMP 120 mg + INH 80 mg+PZA 250 mg)	肝、血液、胃腸不適、 皮膚敏感、高尿酸	口服	1. 成人依體重每增加 10 kg，加服 1 錠，每日最多 5 錠。 2. 每日使用劑量超過五錠的病人，不建議使用 RFT，需單方開立並依體重調整劑量。	
AKuriT-4 (AKT-4)/ Trac 4 tablets (Trac 4) (RMP 150 mg + INH 75 mg + PZA 400 mg + EMB 275 mg)	肝、血液、胃腸不適、皮膚敏感、高尿酸、視神經炎	口服	1. 依體重分級，30-37Kg 者每日口服 2 錠，38-54Kg 者每日 3 錠（50Kg 以上，可考慮依病人肝腎功能調整至每日 4 錠），55-70Kg 者每日 4 錠，≧71Kg 者可考慮投與每日 5 錠。 2. 依體重每日劑量超過 5 錠或體重小於 30 公斤的病人不建議使用 AKT，須單方開立並依體重調整劑量。	
AKuriT-3 (AKT-3)/ Trac 3 tablets (Trac 3) (RMP 150 mg + INH 75 mg + EMB 275 mg)	肝、血液、胃腸不適、皮膚敏感、高尿酸、視神經炎	口服	1. 依體重分級，30-37Kg 者每日口服 2 錠，38-54Kg 者每日 3 錠（50Kg 以上，可考慮依病人肝腎功能調整至每日 4 錠），55-70Kg 者每日 4 錠，≧71Kg 者可考慮投與每日 5 錠。 2. 依體重每日劑量超過 5 錠或體重小於 30 公斤的病人不建議使用 AKT，須單方開立並依體重調整劑量。	

表 6-3 第二線抗結核藥物

藥物(縮寫)	給藥方式	每日建議劑量(範圍)	每日最大劑量	毒性/副作用
kanamycin (KM)	肌肉/靜脈注射	15 mg/kg (15-20)	1000 mg*	耳毒性、腎毒性、暈眩或聽力障礙 孕婦避免
amikacin (AMK)	肌肉/靜脈注射	15 mg/kg (15-20)	1000 mg*	耳毒性、腎毒性、暈眩或聽力障礙
prothionamide (TBN)	口服	15 ~ 20 mg/kg	1000 mg	胃腸不適、肝毒性、低甲狀腺血症、神經/精神系統障礙
para-aminosalicylate (PAS)	口服	150 mg/kg	12 gm	胃腸不適、肝毒性、低甲狀腺血症、皮疹
cycloserine (CS)	口服	10 ~ 15 mg/kg	1000 mg	胃腸不適、神經/精神系統障礙、皮疹
levofloxacin (LFX)	口服	7.5~10 mg/kg (500 ~ 1000 mg)	1000 mg	胃腸不適、頭暈、頭痛、心律不整
moxifloxacin (MFX)	口服	400 mg	400 mg#	胃腸不適、頭暈、頭痛、心律不整

* 累積總劑量不超過 120 gm

\# 800mg is recommended by WHO in 9-month regimen

潛伏結核感染治療處方一覽表

112年印製

處方	處方藥品		總劑數與療程頻率	劑量：每日最大劑量	劑量：兒童	劑量：成人	常見副作用	使用限制	都治(DOPT)	推薦順序(接觸者除指標抗藥或使用限制外)
1HP[a]	複方	Isoniazid(INH) 300mg+ Rifapentine (RPT) 300mg	28天 (1個月) 每日服用	300mg	固定1顆		皮疹(蕁麻疹)為主、(少數)肝毒性	◆ 指標個案INH或RMP抗藥之接觸者 ◆ <13歲兒童 ◆ 孕婦[c]	必須	推薦處方
		Rifapentine (RPT) 150mg		300mg	◆ 35-45 kg 1顆 ◆ >45 kg 2顆					
	單方	Isoniazid (INH) 300mg	28天 (1個月) 每日服用	300mg	300 mg					
		Rifapentine (RPT) 150mg		600mg	◆ <35 kg 300 mg					
3HP[a]	複方	Isoniazid(INH) 300mg+ Rifapentine (RPT) 300mg	12個劑量 (3個月) 每週服用	900 mg						薦處方
	單方	Isoniazid (INH) 300mg	12個劑量 (3個月) 每週服用	900 mg						薦處方
		Rifapentine (RPT) 150mg		900 mg						
4R	Rifampin (RMP) 300mg		120天 (4個月) 每日服用	600 mg	15 (10-20)mg/kg	10 mg/kg	皮疹、腸胃不適/腸胃障礙、(少數)肝毒性	指標個案RMP抗藥之接觸者	必須	推薦處方
3HR[b]	Isoniazid (INH) 100mg		90天 (3個月) 每日服用	300 mg	10 (7-15)mg/kg	5 mg/kg	過敏反應、(少數)肝毒性	指標個案INH或RMP抗藥之接觸者	必須	推薦處方
	Rifampin (RMP) 300mg			600 mg	15 (10-20)mg/kg	10 mg/kg				
6H /9H	Isoniazid(INH) 100mg		180天(6個月) /270天(9個月) 每日服用	300 mg	10 (7-15)mg/kg	5 mg/kg	皮疹、周邊神經病變、肝毒性	指標個案INH抗藥之接觸者	建議	替代處方

Isonizaid每天吃的:300mg
Isoniazid每週吃的:900mg

Rifapentine每天吃的:600mg
Rifapentine每週吃的:900mg

臨床常見劑量錯誤案例-1

- 開立Isoniazid/Rifapentine Tab300mg/300mg 3 #QD。
- 經藥師與臨床端確認後修改為1# QW

臨床常見劑量錯誤案例-2

- Isoniazid/Rifapentine Tab. 300mg/300mg/tab 1# QD-PO 7tab
- 經藥師與臨床端確認後加開Rifapentine 150mg 2# QD

2003年 35%沒有體重記錄，結核藥物的劑量非常不一致
2007 年-2010年一系列措施，2010年，96.5%有體重

OPEN ACCESS Freely available online PLOS | ONE

Improved Consistency in Dosing Anti-Tuberculosis Drugs in Taipei, Taiwan

Chen-Yuan Chiang[1,2,3], Ming-Chih Yu[2,3], Hsiu-Chen Shih[4], Muh-Yong Yen[4], Yu-Ling Hsu[5], Shiang-Lin Yang[5], Tao-Ping Lin[6], Kuan-Jen Bai[2,3]*

1 International Union Against Tuberculosis and Lung Disease, Paris, France, 2 Division of Pulmonary Medicine, Department of Internal Medicine, Wan Fang Hospital, Taipei Medical University, Taipei, Taiwan, 3 Department of Internal Medicine, School of Medicine, College of Medicine, Taipei Medical University, Taipei, Taiwan, 4 Department of Disease Control and Prevention, Taipei City Hospital, Taipei City Government, Taipei, Taiwan, 5 Centers for Disease Control, Department of Health, Taipei, Taiwan, 6 Taiwan Anti-Tuberculosis Association, Taipei, Taiwan

Abstract

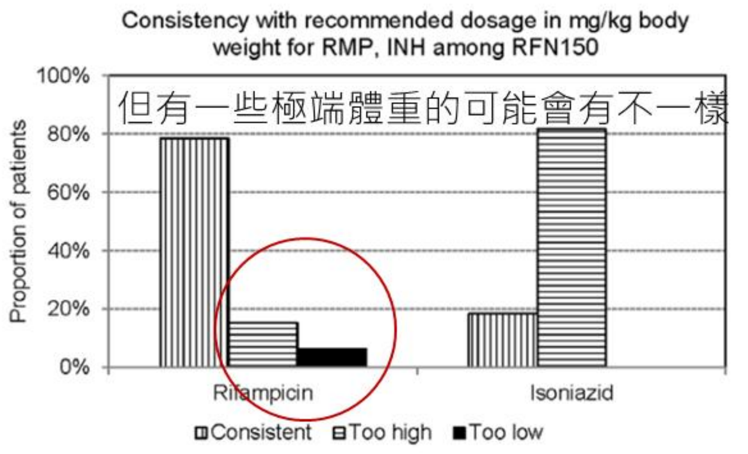
Background: It was reported that 35.5% of tuberculosis (TB) cases reported in 2003 in Taipei City had no recorded pre-treatment body weight and that among those who had, inconsistent dosing of anti-TB drugs was frequent. Taiwan Centers for Disease Control (CDC) have taken actions to strengthen dosing of anti-TB drugs among general practitioners. Prescribing practices of anti-TB drugs in Taipei City in 2007–2010 were investigated to assess whether interventions on dosing were effective.

Methodology/Principal Findings: Lists of all notified culture positive TB cases in 2007–2010 were obtained from National TB Registry at Taiwan CDC. A medical audit of TB case management files was performed to collect pretreatment body weight and regimens prescribed at commencement of treatment. Dosages prescribed were compared with dosages recommended. The proportion of patients with recorded pre-treatment body weight was 64.5% in 2003, which increased to 96.5% in 2007–2010 ($p<0.001$). The proportion of patients treated with consistent dosing of a 3-drug fixed-dose combination (FDC) increased from 73.9% in 2003 to 87.7% in 2007–2010 ($p<0.001$), and that for 2-drug FDC from 76.0% to 86.1% ($p=0.024$), for rifampicin (RMP) from 62.8% to 85.5% ($p<0.001$), and for isoniazid from 87.8% to 95.3% ($p<0.001$). In 2007–2010, among 2917 patients treated with either FDCs or RMP in single-drug preparation, the dosage of RMP was adequate (8–12 mg/kg) in 2571(88.1%) patients, too high in 282(9.7%), too low in 64(2.2%). In multinomial logistic regression models, factors significantly associated with adequate dosage of RMP were body weight and preparations of RMP. Patients weighting <40kg (relative risk ratio (rrr) 6010.5, 95% CI 781.1–46249.7) and patients weighting 40–49 kg (rrr 1495.3, 95% CI 200.6–11144.6) were more likely to receive higher-than-recommended dose of RMP.

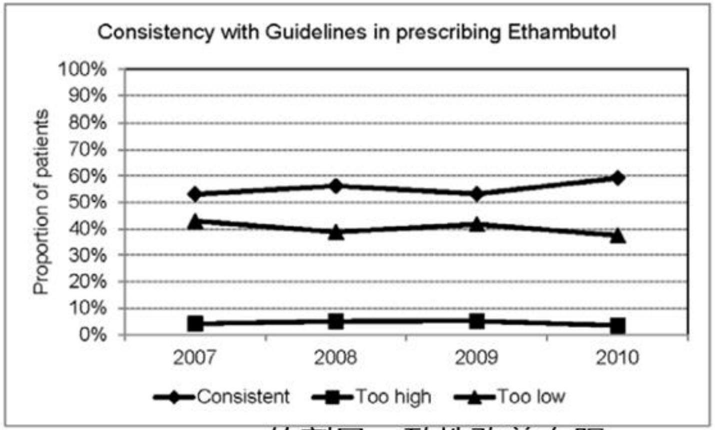
Conclusions/Significance: Prescribing practice in the treatment of TB in Taipei City has remarkably improved after health authorities implemented a series of interventions.

更進階!!!如果可以，調查一下**身高!!!!!!**

使用複方藥物Rifinah150能達80%劑量符合率



但有一些極端體重的可能會有不一樣



Ethambutol 的劑量一致性改善有限，
整體比例從約 54% 微幅增加至 58%，未達統計顯著（**P = 0.136**）

CURRENT OPINION

Therapeutic Drug Monitoring in the Treatment of Tuberculosis: An Update

Abdullah Alsultan · Charles A. Peloquin

濃度偏低與治療失敗相關

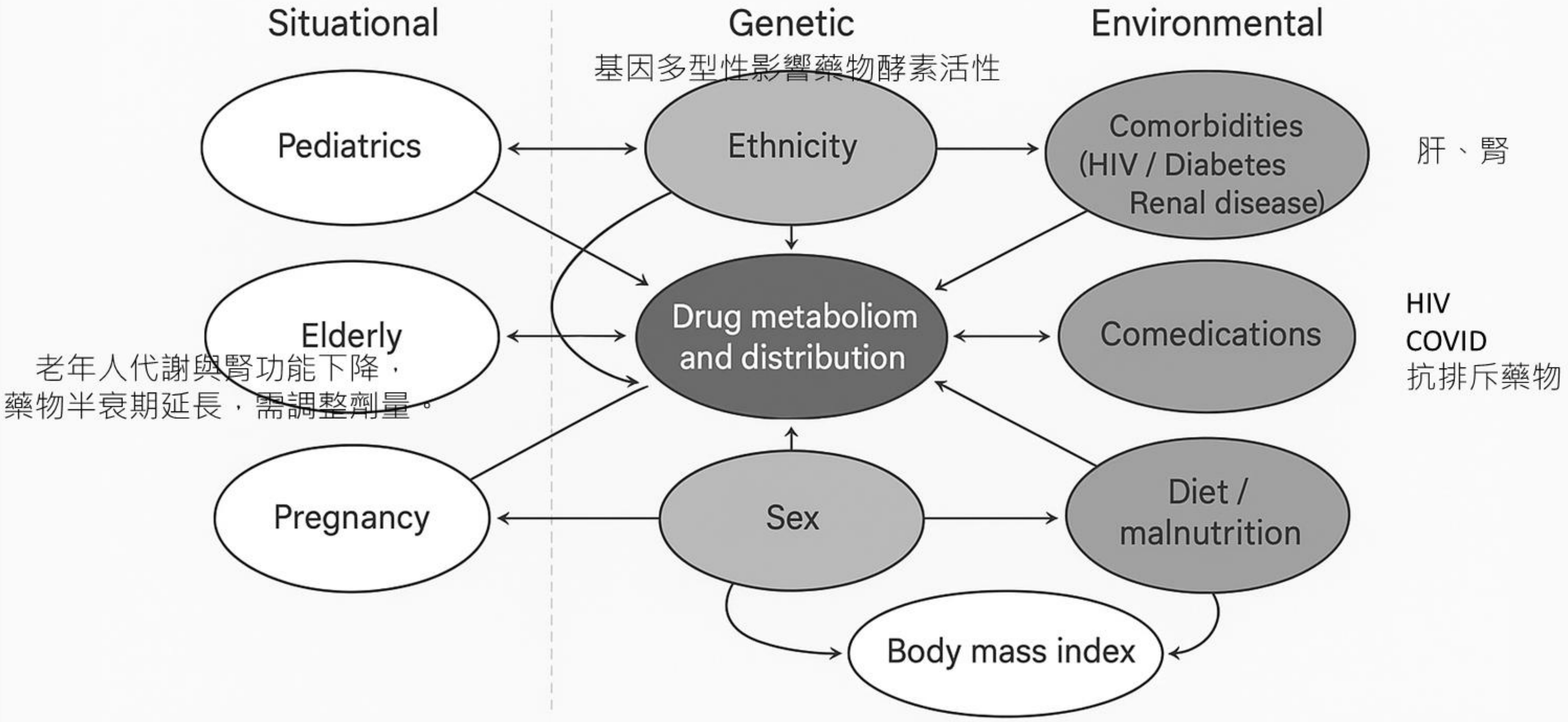
In all three patients who developed drug resistance, rifampicin Cmax concentrations were low... These findings illustrate the importance of drug concentrations as predictors of clinical outcome. 濃度低，造成抗藥性

觀察、等待不是好方法

The alternative to TDM is to watch, wait, and hope for the best. It is common for patients to have their treatment regimens empirically extended when responses to treatment are delayed... prolonged treatment means prolonged expenditure, and failures and relapses mean considerably larger expenditures.

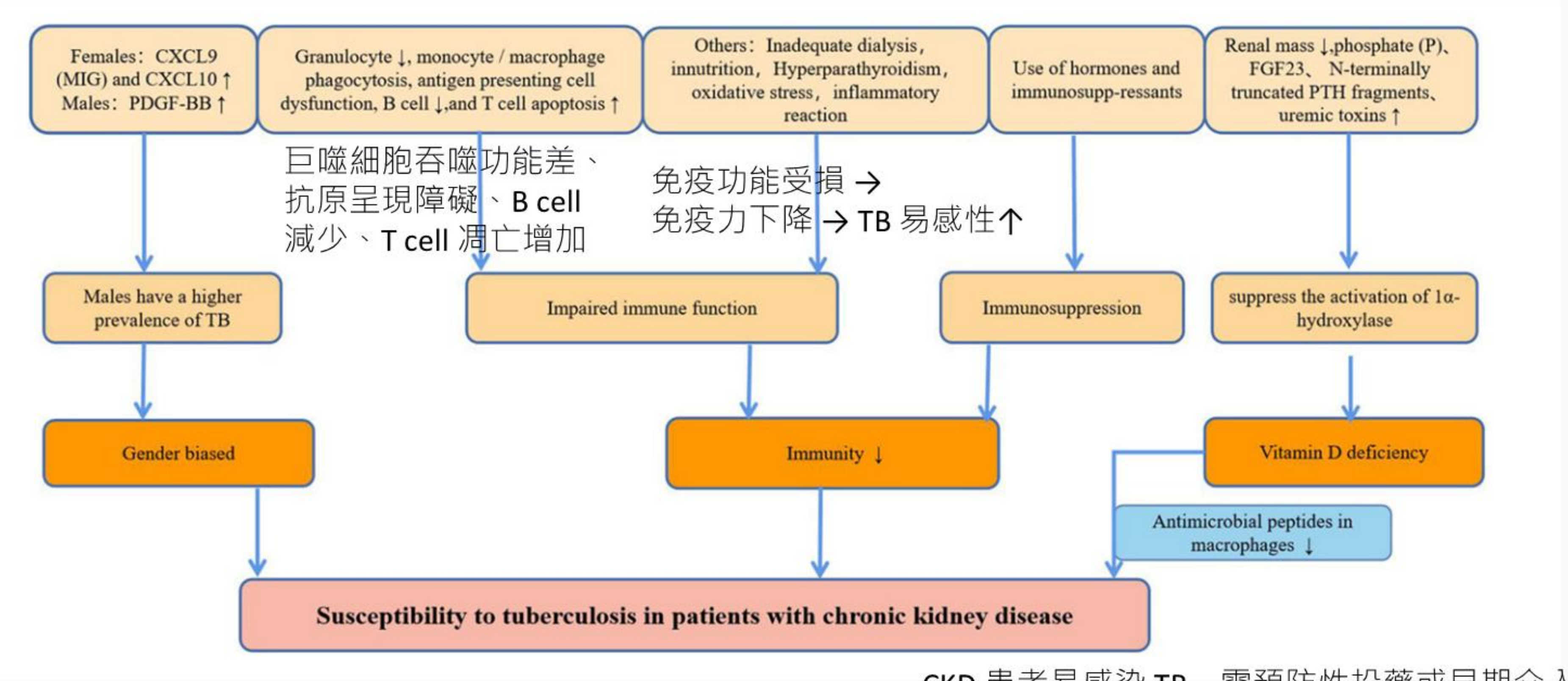
須注意共病或交互作用

Patients with HIV infection... may have various forms of malabsorption... Intermittent dosing and low drug concentrations can be particularly dangerous in HIV-positive patients



J Infect Dis. 2015 Jun 15;211 Suppl 3(Suppl 3):S115-25.

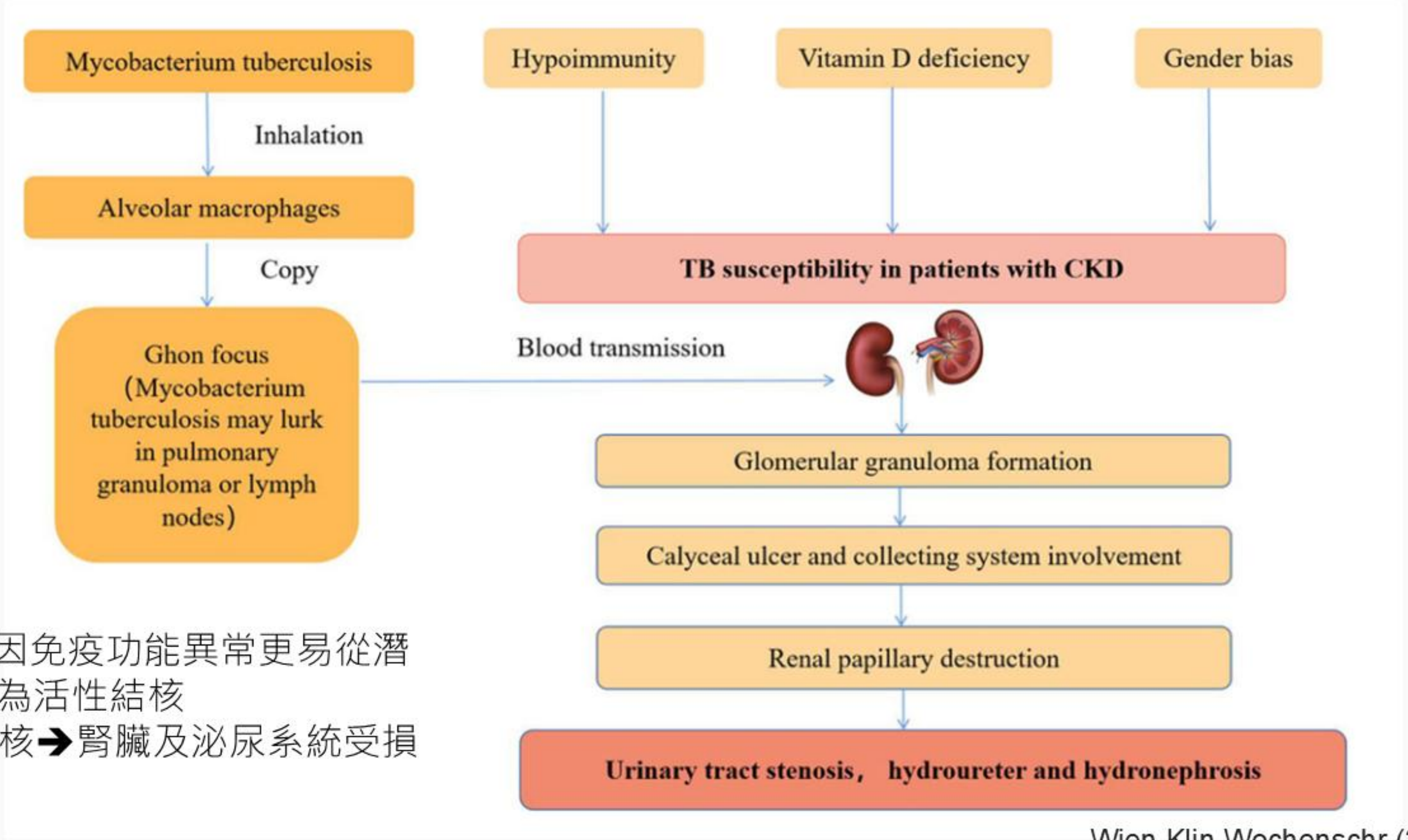
腎功能不好對TB具易感性



CKD 患者易感染 TB，需預防性投藥或早期介入。

Wien Klin Wochenschr (2022) 134:501–510

TB也會造成CKD病人腎損傷



CKD 患者因免疫功能異常更易從潛伏結核轉為活性結核
➔泌尿結核➔腎臟及泌尿系統受損

. Wien Klin Wochenschr (2022) 134:501–510

Effect of chronic kidney disease on adverse drug reactions to anti-tubercular treatment: a retrospective cohort study

印度回溯性世代研究，把要接受抗結核治療的病人分為CKD跟Normal kidney function 1815人(75:1740)

Divya Datta, Indu Ramachandra Rao, Attur Ravindra Prabhu, Shankar Prasad Nagaraju, Girish Thunga, Rahul Magazine, Shivashankar Kaniyoor Nagri, Raghavendra Shetty, Nisha Abdul Khader, Dharshan Rangaswamy, Srinivas Vinayak Shenoy, Mohan V. Bhojaraja & Asha Kamath

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To link to this article: https://doi.org/10.1080/0886022X.2024.2392883

Ren Fail. 2024 Dec;46(2):2392883.

CKD 與非 CKD 患者在藥物不良反應有差異

Table 3. ATT-associated adverse drug reactions in the study population.

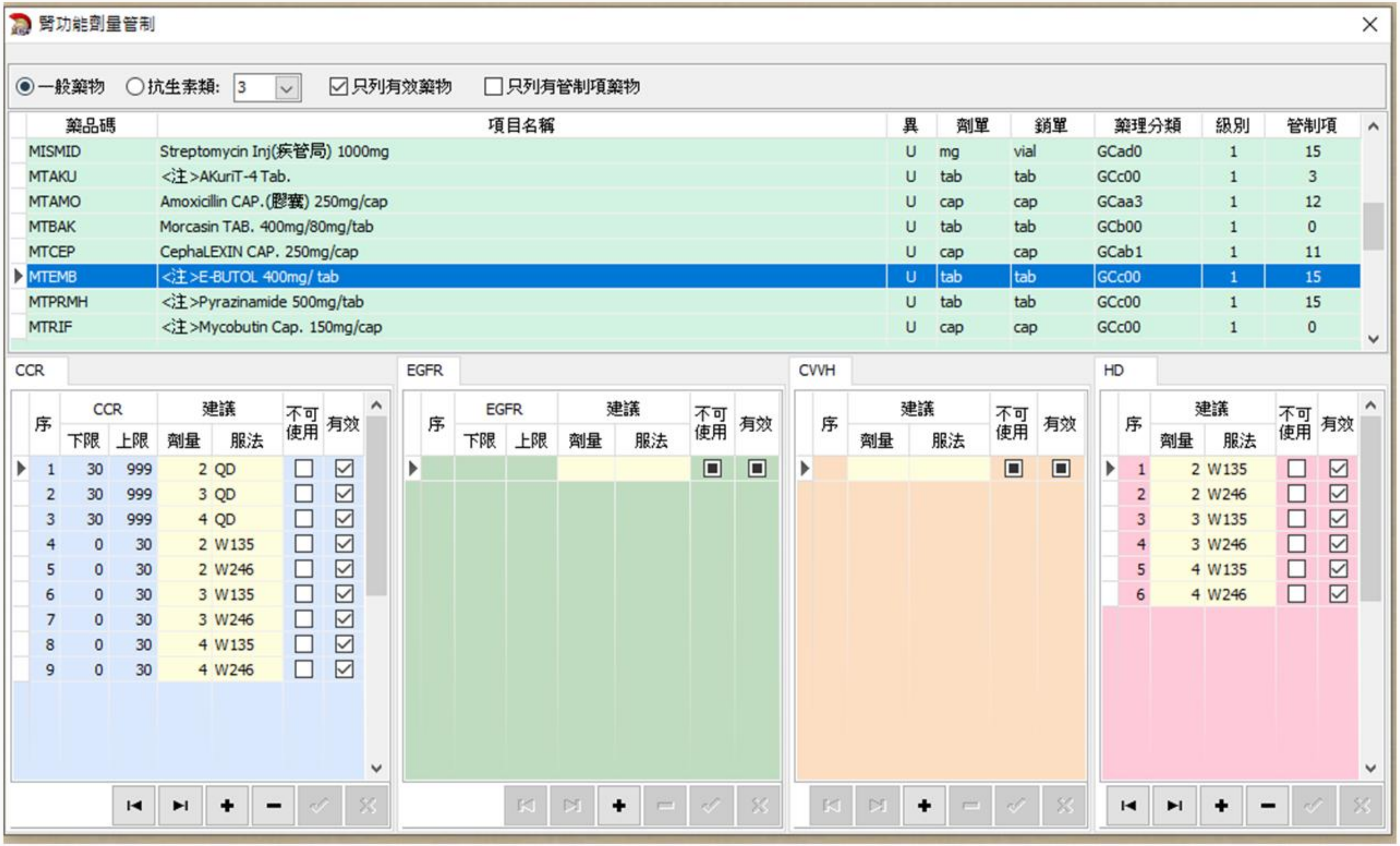
Adverse drug reactions	Total ($n=1815$)	CKD ($n=75$)	Normal kidney function ($n=1740$)	*P*-value
Number of patients with adverse drug reactions	275 (15.1%)	36 (48%)	239 (13.7%)	**<0.0001**
Number of adverse reactions observed	329 (18.1%)	46 (61.3%)	283 (16.2%)	**<0.0001**
Type of ADR				
Drug-induced hepatitis 藥物引起的肝炎	179 (9.8%)	23 (30.6%)	156 (8.9%)	**<0.0001**
Gastrointestinal (nausea, vomiting)	39 (2.1%)	5 (6.6%)	34 (1.9%)	**0.020**
Dermatological (acneiform eruption, dermatitis, DRESS, maculopapular rash, pellagrous dermatitis, urticaria, xerosis) 皮疹	32 (1.7%)	1 (1.3%)	31 (1.7%)	1.000
Neurological and neuropsychiatric (psychosis, encephalopathy, seizures, peripheral neuropathy, optic neuropathy) 神經學的病變	29 (1.5%)	8 (10.6%)	21 (1.2%)	**<0.0001**
Arthritis	12 (0.6%)	1 (1.3%)	11 (0.6%)	0.398
Acute kidney injury 腎功能	12 (0.6%)	4 (5.3%)	8 (0.4%)	**0.001**
Ear (high-frequency hearing loss, vestibular toxicity)	9 (0.4%)	1 (1.3%)	8 (0.4%)	0.317
Hyperuricemia	7 (0.3%)	1 (1.3%)	6 (0.3%)	0.256
Hematological (coagulopathy, pancytopenia, hemolytic anemia, thrombocytopenia)	6 (0.3%)	1 (1.3%)	5 (0.2%)	0.224
Thyroiditis	4 (0.2%)	1 (1.3%)	3 (0.1%)	0.155

ADR: adverse drug reactions; CKD: chronic kidney disease; DRESS: drug reaction with eosinophilia and systemic symptoms.
P-values which came significant have been made bold.

Ren Fail. 2024 Dec;46(2):2392883.

腎功能不全病人

藥物名稱	標準劑量	eGFR <30	HD	PD	CRRT	備註
Isoniazid (INH)	5 mg/kg 每日（最大 300 mg）	不需調整	透析後給藥	不需調整	不需調整	少量透析清除，建議補 B6
Rifampin (RMP)	10 mg/kg 每日（最大 600 mg）	不需調整	透析後給藥	不需調整	不需調整	肝代謝，不透析
Pyrazinamide (PZA)	25–35 mg/kg 每日	每週 3 次	每週 3 次，透析後給藥	建議延長間隔	每 24–48 小時給 25–35 mg/kg	代謝物蓄積風險高
Ethambutol (EMB)	15–25 mg/kg 每日	每週 3 次	每週 3 次 透析後給藥	每週 3 次	每日或隔日給藥	需監測視神經毒性
Streptomycin (SM)	12–15 mg/kg 每日	每週 2–3 次	每週 2–3 次 透析後給藥	慎用（PD 清除差）	每 48–72 小時給藥	耳毒性、腎毒性，建議監測濃度
Levofloxacin	750–1000 mg 每日	每週 3 次	每週 3 次 透析後給藥	每週 3 次	不需調整	腎排除，建議調整頻次
Moxifloxacin	400 mg 每日	不需調整	可於透析後給藥	不需調整	不需調整	肝代謝，可使用
Cycloserine	250–500 mg 每日	每週 3 次或減半	每週 3 次 透析後給藥	慎用，建議延長間隔	劑量調整需監測	神經毒性高，需警覺
Ethionamide	250 mg BID	不需調整	可於透析後給藥	不需調整	不需調整	胃腸副作用常見
PAS	2 g BID	減量	透析後給藥	減量	需調整	PAS + EMB 增加甲狀腺低下風險



腎功能劑量管制

◉一般藥物 ○抗生素類: 3 ☑只列有效藥物 ☐只列有管制項藥物

藥品碼	項目名稱	異	劑單	銷單	藥理分類	級別	管制項
MISMID	Streptomycin Inj(疾管局) 1000mg	U	mg	vial	GCad0	1	15
MTAKU	<注>AKuriT-4 Tab.	U	tab	tab	GCc00	1	3
MTAMO	Amoxicillin CAP.(膠囊) 250mg/cap	U	cap	cap	GCaa3	1	12
MTBAK	Morcasin TAB. 400mg/80mg/tab	U	tab	tab	GCb00	1	0
MTCEP	CephaLEXIN CAP. 250mg/cap	U	cap	cap	GCab1	1	11
MTEMB	<注>E-BUTOL 400mg/ tab	U	tab	tab	GCc00	1	15
MTPRMH	<注>Pyrazinamide 500mg/tab	U	tab	tab	GCc00	1	15
MTRIF	<注>Mycobutin Cap. 150mg/cap	U	cap	cap	GCc00	1	0

CCR

序	CCR 下限	CCR 上限	建議 劑量	建議 服法	不可使用	有效
1	30	999	2	QD	☐	☑
2	30	999	3	QD	☐	☑
3	30	999	4	QD	☐	☑
4	0	30	2	W135	☐	☑
5	0	30	2	W246	☐	☑
6	0	30	3	W135	☐	☑
7	0	30	3	W246	☐	☑
8	0	30	4	W135	☐	☑
9	0	30	4	W246	☐	☑

EGFR

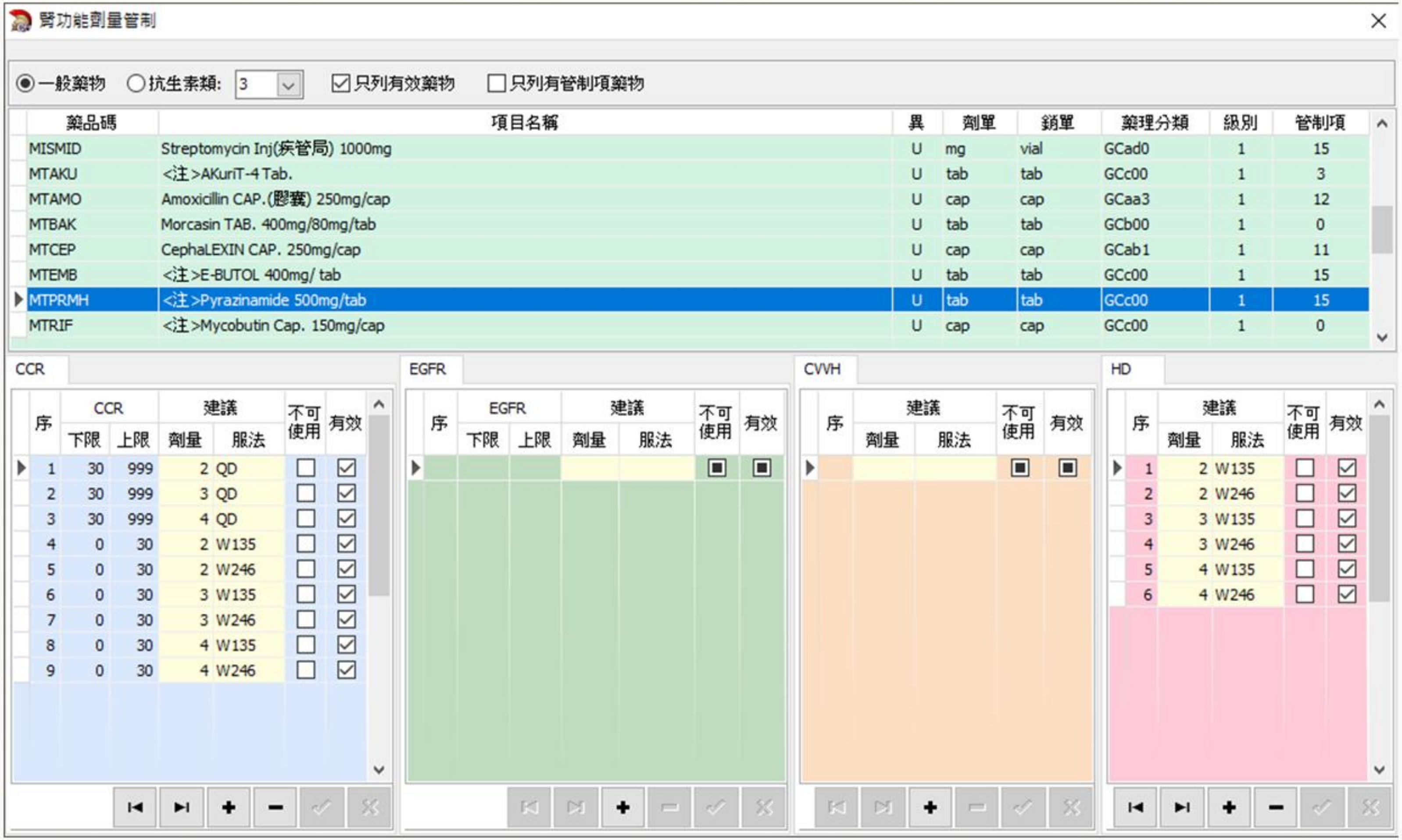
序	EGFR 下限	EGFR 上限	建議 劑量	建議 服法	不可使用	有效

CVVH

序	建議 劑量	建議 服法	不可使用	有效

HD

序	建議 劑量	建議 服法	不可使用	有效
1	2	W135	☐	☑
2	2	W246	☐	☑
3	3	W135	☐	☑
4	3	W246	☐	☑
5	4	W135	☐	☑
6	4	W246	☐	☑


腎功能劑量管制

◉一般藥物 ○抗生素類: 3 ☑只列有效藥物 ☐只列有管制項藥物

藥品碼	項目名稱	異	劑單	銷單	藥理分類	級別	管制項
MISMID	Streptomycin Inj(疾管局) 1000mg	U	mg	vial	GCad0	1	15
MTAKU	<注>AKuriT-4 Tab.	U	tab	tab	GCc00	1	3
MTAMO	Amoxicillin CAP.(膠囊) 250mg/cap	U	cap	cap	GCaa3	1	12
MTBAK	Morcasin TAB. 400mg/80mg/tab	U	tab	tab	GCb00	1	0
MTCEP	CephaLEXIN CAP. 250mg/cap	U	cap	cap	GCab1	1	11
MTEMB	<注>E-BUTOL 400mg/ tab	U	tab	tab	GCc00	1	15
MTPRMH	<注>Pyrazinamide 500mg/tab	U	tab	tab	GCc00	1	15
MTRIF	<注>Mycobutin Cap. 150mg/cap	U	cap	cap	GCc00	1	0

CCR

序	CCR 下限	CCR 上限	建議 劑量	建議 服法	不可使用	有效
1	30	999	2	QD	☐	☑
2	30	999	3	QD	☐	☑
3	30	999	4	QD	☐	☑
4	0	30	2	W135	☐	☑
5	0	30	2	W246	☐	☑
6	0	30	3	W135	☐	☑
7	0	30	3	W246	☐	☑
8	0	30	4	W135	☐	☑
9	0	30	4	W246	☐	☑

EGFR

序	EGFR 下限	EGFR 上限	建議 劑量	建議 服法	不可使用	有效

CVVH

序	建議 劑量	建議 服法	不可使用	有效

HD

序	建議 劑量	建議 服法	不可使用	有效
1	2	W135	☐	☑
2	2	W246	☐	☑
3	3	W135	☐	☑
4	3	W246	☐	☑
5	4	W135	☐	☑
6	4	W246	☐	☑

香港3007位Retrospective Cohort Study、Nested Case-Control Study

Pyrazinamide肝毒性風險較高

Multivariable conditional logistic regression

抗結核治療超過 12 週後哪些因素會顯著增加肝毒性的風險

TABLE 3. MULTIVARIABLE CONDITIONAL LOGISTIC REGRESSION ANALYSIS OF HEPATOTOXICITY FROM 12 OR MORE WEEKS AFTER STARTING TREATMENT INVOLVING 47 CASES AND 141 MATCHED CONTROL SUBJECTS

Explanatory Variables	OR (*95% CI*)	*P*
Predominant regimens received within 4 weeks preceding hepatotoxicity, with reference to regimens comprising H and R		0.006
Pyrazinamide-containing regimens: HRZ or HZ or RZ	2.8 (1.4–5.9)	0.005
Other regimens	2.7 (1.1–6.8)	0.04
Previous hepatotoxicity (after treatment < 12 wk)	2.8 (1.1–7.0)	0.03
Hepatitis B	2.1 (1.1–4.1)	0.03

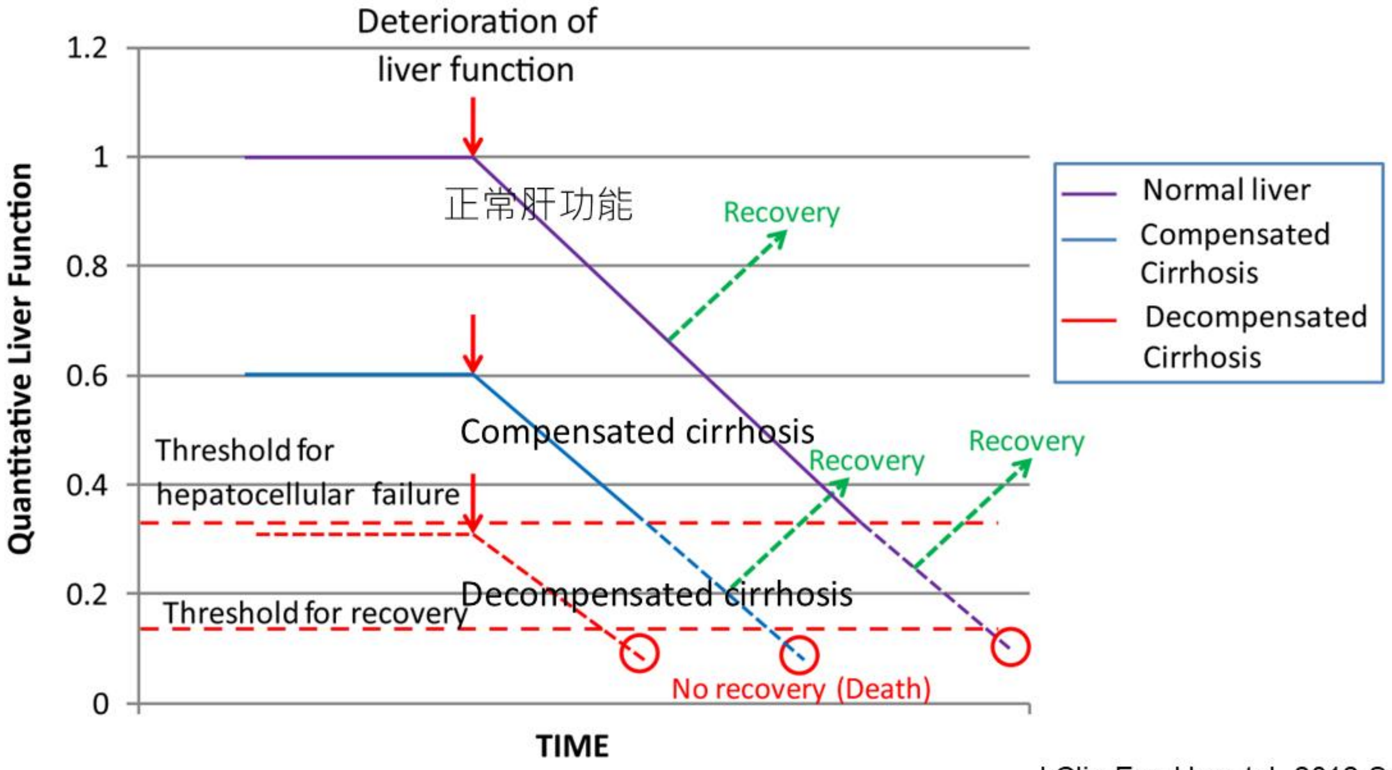
Pyrazinamide 的組合會讓肝毒性風險提高 2.8 倍

之前有肝炎、或曾有肝毒性經驗

Definition of abbreviations: CI = confidence interval; H = isoniazid; OR = odds ratio; R = rifampin; Z = pyrazinamide.

Am J Respir Crit Care Med. 2008 Jun 15;177(12):1391-6.

肝功能差的人殺傷力更大，且更難恢復



J Clin Exp Hepatol. 2012 Sep 21;2(3):260–270.

肝功能不全的病人

藥物名稱	肝毒性	肝功能不全時建議	備註
INH (isoniazid)	肝毒性次強 是	重啟第 2 位順序	slow acetylator 劑量需調整
RIF (rifampicin)	是 肝毒性第三強	優先重啟 用於膽汁鬱積型慎用	較低肝毒性
PZA (pyrazinamide)	肝毒性最強 高	建議避免/最後加/慎用	重度肝病建議完全停用
EMB (ethambutol)	否	安全，可維持基本殺菌力	建議第一時間 納入使用
Levo/Moxifloxacin 也不是完全不會	否	安全，重度肝病首選替代藥物	

WHO:加藥回來的順序RIF➔INH➔PZA

WHO operational handbook on tuberculosis – Module 4: Treatment – drug-susceptible tuberculosis treatment, 2022 update.

Child-Pugh應用於藥物選擇:依賴臨床判斷

Child-Turcotte-Pugh score	Liver disease	Treatment
≤7	Stable	Recommend treatment with two potentially hepatotoxic drugs, likely to be well tolerated; avoid pyrazinamide
8–10	Advanced	Recommend a regimen with only one potentially hepatotoxic drug; rifampicin is preferred over isoniazid; pyrazinamide should not be used
≥11	Very advanced	Recommend treatment regimen with no potentially hepatotoxic drugs; can use (streptomycin, ethambutol, fluoroquinolones, amikacin, kanamycin) and other second-line oral drugs for 18–24 months.

用第二線藥物但治療比較久

	1 point	2 points	3 points
Ascitis	None	Mild	Moderate to severe
Serum albumin (g/dl)	>3.5	2.8–3.5	<2.8
Bilirubin, total (mg/dl)	<2	2–3	>3
Hepatic encephalopathy	No	Grade I–II	Grade III–IV
Prothrombin time (INR)	<1.7	1.7–2.3	>2.3

J Clin Exp Hepatol. 2012 Sep 21;2(3):260–270.

肝功能<5倍且沒症狀
肝功能<3倍且症狀輕微
可先追蹤

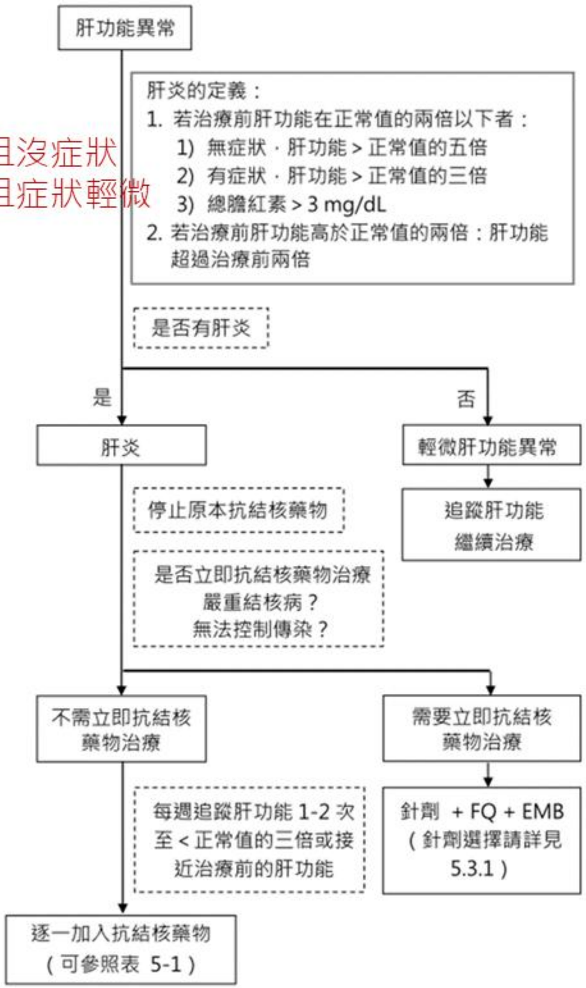


表 5-1 藥物性肝炎小量漸進式給藥試驗流程建議

日期(天)	藥物	劑量	肝功能檢測 [a]
0	-	-	+
1	INH	100 mg/day	
2	INH	200 mg/day	
3~5	INH	full dose	+
6	+RMP	150 mg/day	
7	+RMP	300 mg/day	
8~10	+RMP	full dose	+
11	+PZA	250 mg/day	
12	+PZA	500 mg/day	
13	+PZA	full dose	+

a：建議包含 ALT、AST、以及 total bilirubin。
逐一加藥的過程中，可以同時使用足夠劑量的EMB。
若病人之藥物性肝炎嚴重或合併有黃疸，在成功地重新使用上INH及RMP之後，不建議嘗試加入PZA。

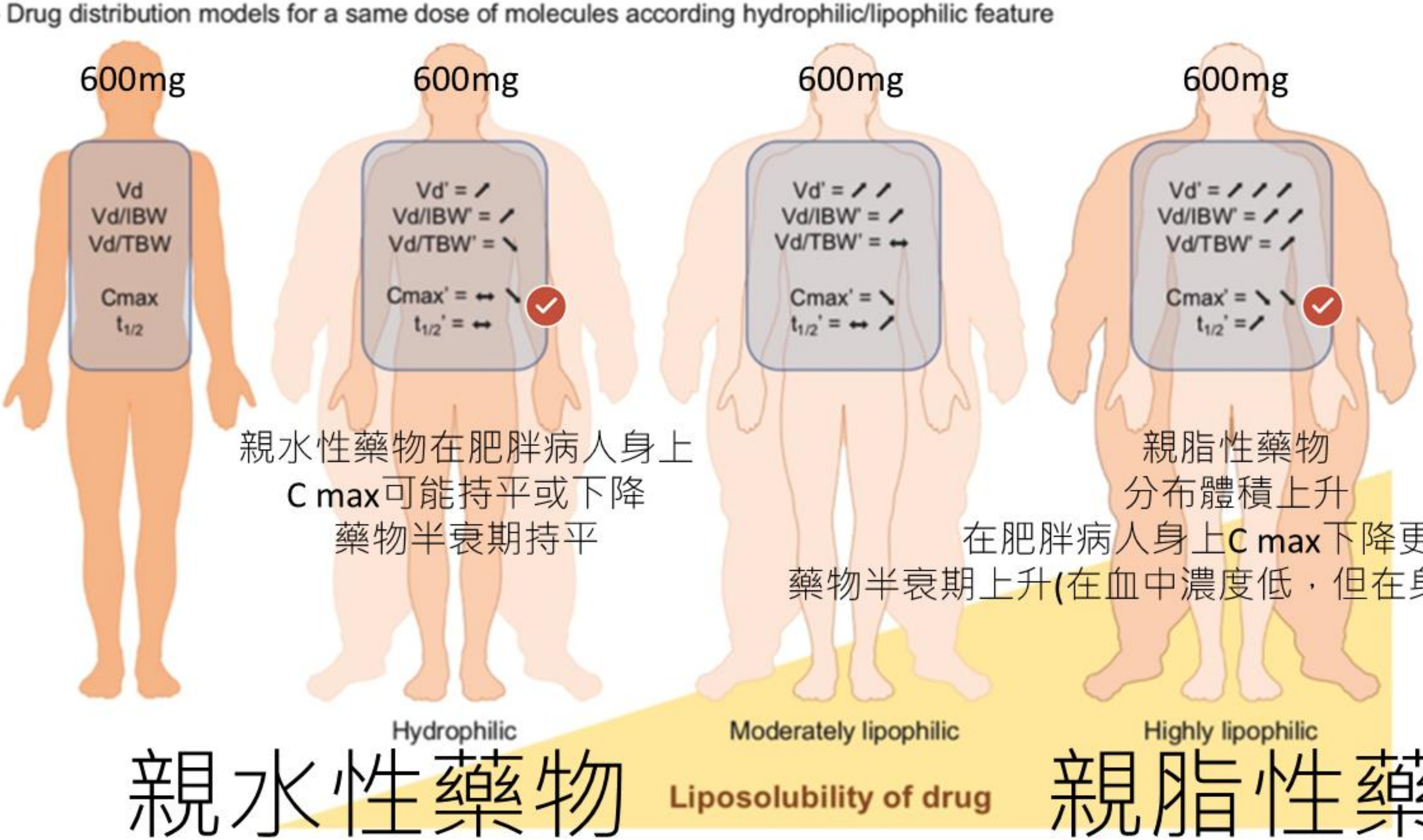
肝功能異常處理不一致-但概念類似

來源	用藥建議 INH有早期殺菌力 為了保護RMP，避免抗藥	Child-Pugh
台灣《結核病診治指引第七版》	說明 DILI 處理與加藥順序沒有 **Child** 分級選藥表 先加回**INH**、再加回**RMP**、最後再加**PZA**	無
WHO Operational Handbook 2022	建議避免肝毒性藥物、採用 EMB/FQ	無
ATS/CDC 2016（美國）	強調監測與「個別化處置」	無
Dhiman 2012 （A Guide to TB in CLD）	明列 Child A/B/C 對應藥物方案	有

相對於腎功能-肝功能介入系統不好輔助

臨床藥師
can help

一樣劑量，在肥胖者可能藥效不夠



親水性藥物在肥胖病人身上
C max可能持平或下降
藥物半衰期持平

親脂性藥物
分布體積上升
在肥胖病人身上C max下降更顯著
藥物半衰期上升(在血中濃度低，但在身體時間停留長)

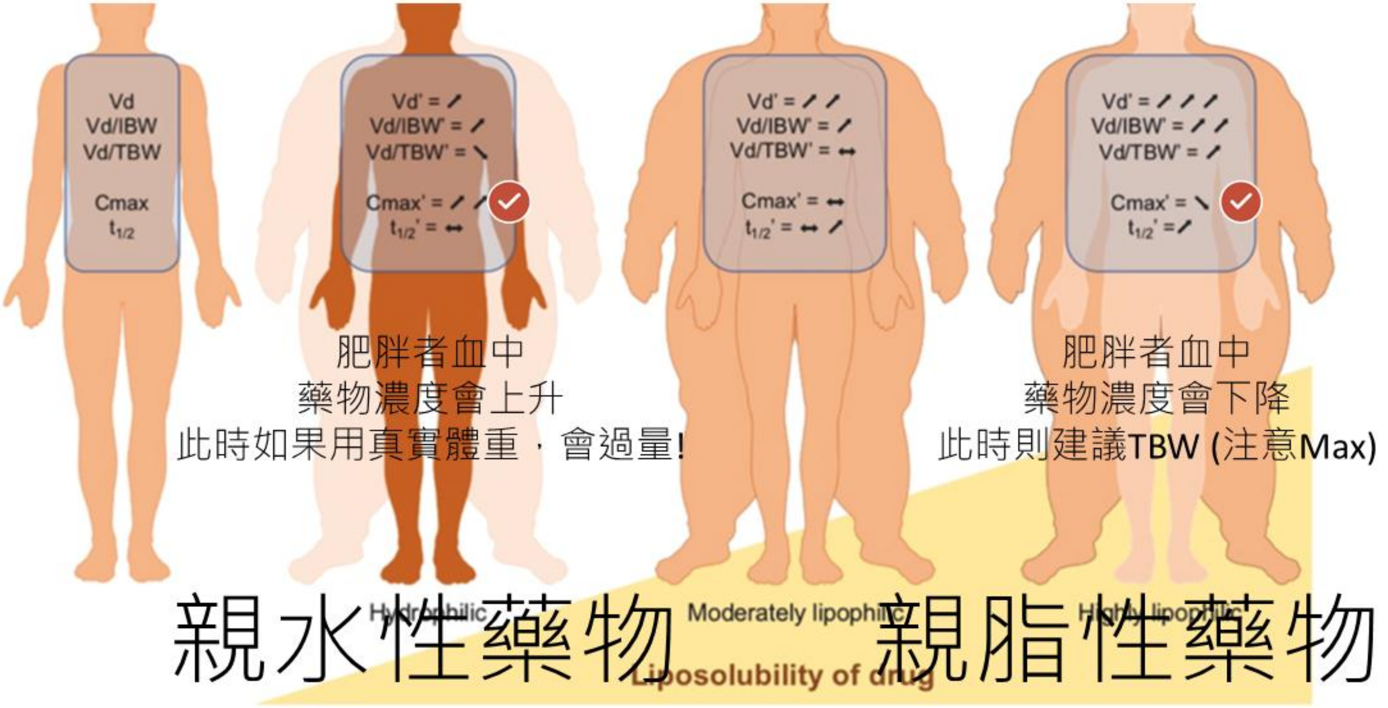
親水性藥物

親脂性藥物

Biomed Pharmacother. 2023 Oct:166:115281.

親水性藥物使用IBW或ABW，親脂性TBW

使用Total body weight去計算劑量



肥胖者血中
藥物濃度會上升
此時如果用真實體重，會過量!

肥胖者血中
藥物濃度會下降
此時則建議TBW (注意Max)

親水性藥物　　親脂性藥物

Biomed Pharmacother. 2023 Oct:166:115281.

臨床案例:72.6 y/o Female 160cm 102.2kg

2025-06-29 Creatinine:0.87 mg/dL　2025-06-29 CrCl:66 ml/min　2025-06-29 Estimated GFR:64.00 mL/min/1.73　2025-06-29 GPT (ALT):20 U/L
2025-06-29 I.N.R.:0.99　2025-06-30 Na:140 mmol/L　2025-06-30 K:4.1 mmol/L　2025-06-30 Ca:8.0 mg/dL
2025-06-30 Mg:1.9 mg/dL　2025-06-30 P:3.3 mg/dL　2025-06-29 HbA1c:10.4 %　2025-06-29 Albumin:3.4 g/dL

基本資料(一) | 基本資料(二) | 訪視記錄資料 | 檢驗報告 | MI檢驗報告 | 藥　囑 | 藥　歷 | 雲端藥歷 | ADR/G6PD | 照會藥師 | 審核記錄 | 重點藥品檢視 | 藥囑警示 | 抗血栓藥物復藥

現況常規及三日Stat藥品醫囑： 中藥處方匯入現況用藥　伙食醫囑：　門診醫囑 查詢　系統審核

生效時間	抗	重	藥品名稱	劑量服法	途徑	停用時間	備註
2025-06-29 21:35			Insulin Glargine SoloStar(筆型) 450 IU/1.5mL/pen (Toujeo (筆型))	14 IU HS	SC		自備藥
2025-06-29 21:35			【橘】Insulin Aspart FlexPen(超短效) 300IU/3mL (【橘】NovoRapid FlexPen(超短效))	12 IU TID	SC		自備藥
2025-06-29 22:39			Lorazepam INJ 2mg/1mL (AnxiCAM INJ★由護理人員領回★)	2 mg Q6H,Prn	IB		@PRN使用原因:if seizure
2025-06-29 23:35			Enoxaparin 60mg/6000IU/0.6mL/syr (Clexane INJ)	60 mg QD,ST 60.0	SC		
2025-06-30 10:22			【0.9% 500mL】Sodium Chloride INJ(Normal Saline INJ)	500 mL QD	IB		for Veklury
2025-06-30 10:22			【0.9% 500mL】Sodium Chloride INJ(Normal Saline INJ)	500 mL QD	IV		for A-line
2025-06-30 10:22			【0.9% 500mL】Sodium Chloride INJ(Normal Saline INJ)	1000 mL QD	IVDP		每瓶(袋)輸注24小時
2025-07-01 07:37			Sennosides 12.5mg/tab (Sennapur)	2 tab HS	PO		
2025-07-01 09:34			【橘】Insulin Aspart FlexPen(超短效) 300IU/3mL (【橘】NovoRapid FlexPen(超短效))	1 IU ASO,Prn	SC		@PRN使用原因:aso
2025-07-01 13:20	抗		Acyclovir INJ. 250mg (Zovirax INJ.)	900 mg Q8H,ST 900.0	IB	2025-07-05 13:20	
2025-07-01 15:40			【500mg/vial】Methylprednisolone INJ(Solu-medrol INJ)	500 mg Q12H	IVDP	2025-07-06 15:40	1. 500mg solu-mderol + N/S 250ml
2025-07-01 15:40			【0.9% 250mL】0.9% Sodium Chloride 250ml(0.9% NaCl)	250 mL QID	IVD	2025-07-06 15:40	1. 500mg solu-mderol + N/S 250ml

生效時間	抗	重	藥品名稱	劑量服法	途徑	停用時間	備註
2025-06-29 11:52			Lorazepam INJ 2mg/1mL (AnxiCAM INJ★由護理人員領回★)	ST 4	IB		
2025-06-29 12:06			Magnesium Sulfate Inj. 10%;2g/20mL(鎂16.23mEq) (Magnesium Sulfate Inj.)	ST 2	IB		
2025-06-29 12:23			【500mL】Ringer's Lactate Sol'n(Hartmann's)	ST 500	IVD		
2025-06-29 13:22			【0.9% 500mL】Sodium Chloride INJ(Normal Saline INJ)	ST 500	IVD		
2025-06-29 14:30			【500mL】Ringer's Lactate Sol'n(Hartmann's)	ST 500	IVD		
2025-06-29 14:30			【0.9% 500mL】Sodium Chloride INJ(Normal Saline INJ)	ST 500	IVD		
2025-06-29 20:12			【500mL】Ringer's Lactate Sol'n(Hartmann's)	ST 500	IVD		每瓶(袋)輸注24小時
2025-06-29 20:19			AcetaMINOPHEN TAB. 500mg/tab (Acetal TAB.)	ST 1	PO		
2025-06-29 21:33			Humulin R INJ 1000 IU/10mL	ST 6	SC		
2025-06-30 12:39			Calcium Chloride Inj 400mg/20mL (Vitacal Inj.)	ST 400	IB		

藥理分類 Category	ANTIMICROBIAL AGENT; Antiviral Agents; Anti-Herpetic
學名 Generic name	Acyclovir Inj
商品名 Trade name	Zovirax INJ.
藥品碼 Code	MIACV 藥品外觀描述 Appearance 藥品交互作用 Drug interaction 注射藥品須知 Injectable drugs
使用院區 LOC	總院、雲基、鹿基、佑民、二基
規格/劑量單位/健保價 Supply/Units/NHI Price	250mg/vial ＄196
用法用量 Dosage	I.V. infusion over 1 hr; (1)Mucosal & Cutaneous HSV Infections: Adult, 5 mg/kg q8h for 7 days. Infants & children, 10 mg/kg q8h for 10 days. (2)HSV Encephalitis: Adult, 10 mg/kg q8h for 10 days. Children 3 months-12 yrs, 20 mg/kg q8h for 10 days. (3)Varicella Zoster Infections & CMV Prophylaxis in Immunocompromised Host: Adult, 10 mg/kg q8h for 7 days. (4)HSV Encephalitis in Immunocompromised Host: Children, 3 months-12 yrs, 500 mg/m2 q8h. (5)Hemopoietic stem cell transplant - Herpes simplex; Prophylaxis: Adults/adolescents (> 40 kg), 250 mg/m2 q12h or 5 mg/kg q8h. Children (< 40 kg), 250 mg/m2 q8h or 125 mg/m2 q6h (max. 80 mg/kg/day).

Obesity Dosing Adjustments

by Douglas Black, Pharm.D. last updated Jul 1, 2025

Obesity Dosing Adjustments

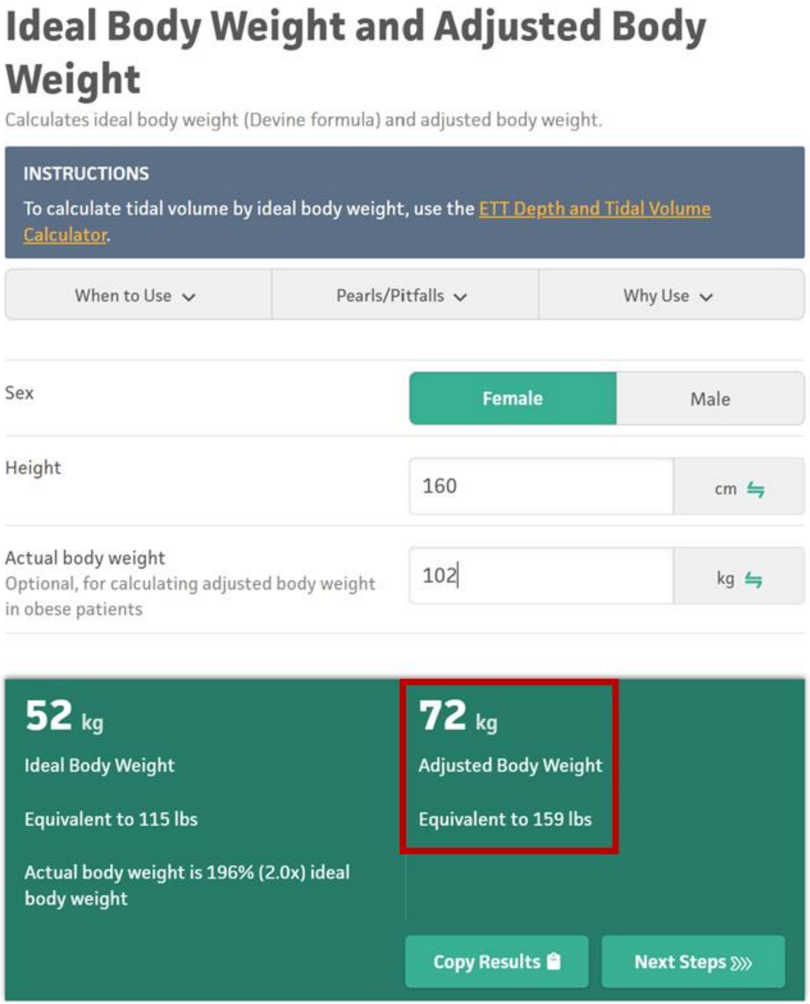
Introduction

- The number of obese patients is increasing. Intuitively, the **standard doses of some drugs may not achieve effective serum concentrations**. Pertinent data on anti-infective dosing in the obese patient are gradually emerging. Even though some of the data need further validation, the following table reflects what is currently known.
- Obesity is defined as weight ≥20% over ideal body weight (Ideal BW) or body mass index (BMI) >30.
- Click for calculator: Ideal body weight (Ideal BW); body mass index (BMI); lean body weight (LBW)
- The pharmacokinetics of antibacterials in obese patients is emerging, but only some drugs have been evaluated. Those drugs for which data justifies a dose adjustment are summarized under Obesity Dosing Adjustment (table below).

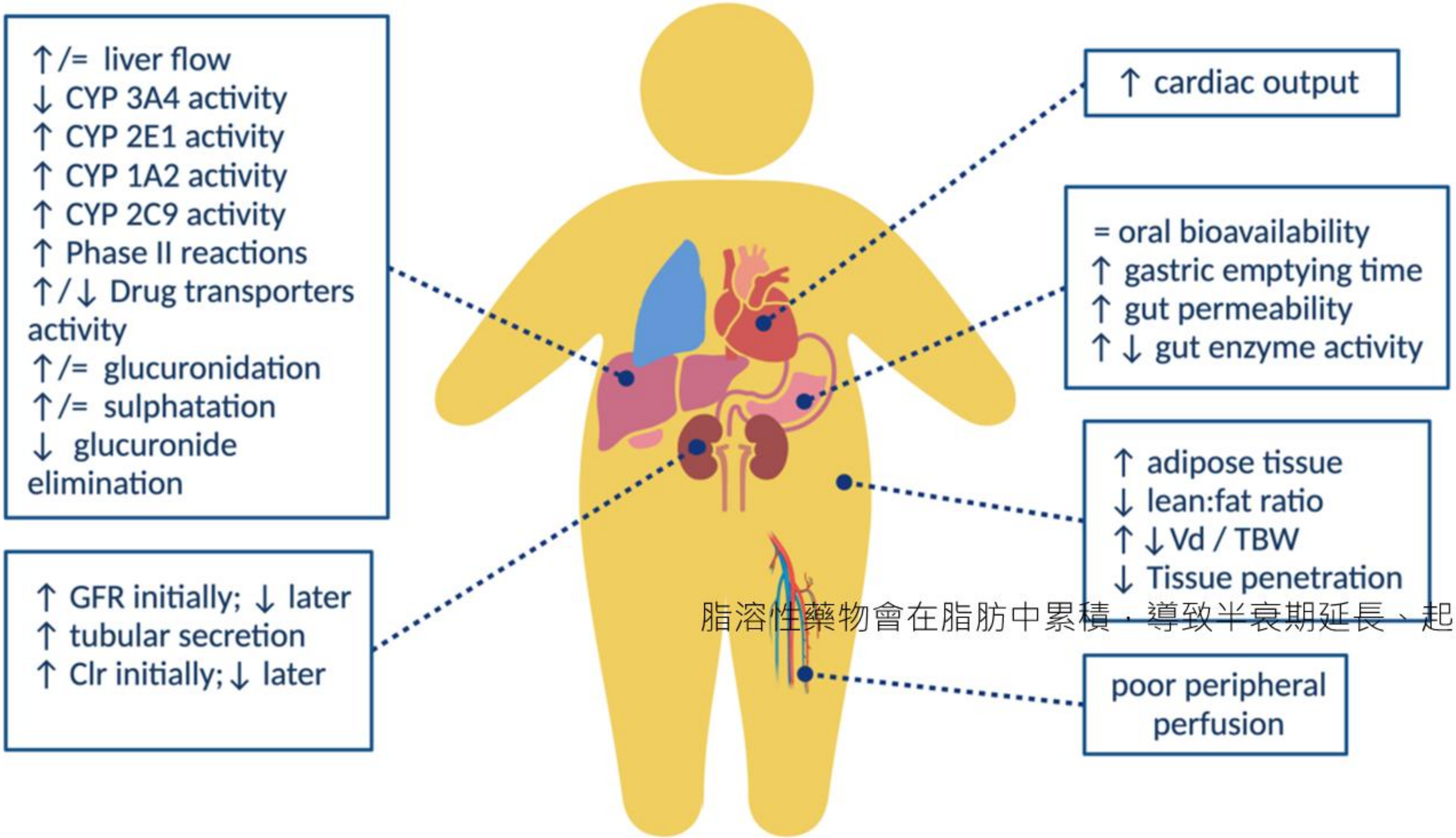
Obesity Dosing Adjustments

- Click drug for full information.
- Dose = suggested body weight (Actual, Ideal, or Adjusted body weight) for dose calculation in an obese patient, or specific dose if applicable. Adjusted BW = Ideal BW + 0.4 x (Actual BW – Ideal BW).

Drug	Recommendation	Comments
Acyclovir	$\boxed{\text{Use Adjusted BW}}$	PK data suggest that adjusted BW in obesity approximates drug exposure seen in non-obesity using actual BW (Antimicrob Agents Chemother 2016;60:1830). Use of adjusted BW in obesity also supported by a retrospective observational study (J Clin Neurosci 2025;136:111230).



劑量從900mg Q8H
調整成750mg Q8H



脂溶性藥物會在脂肪中累積，導致半衰期延長、起效變慢。

Antimicrob Agents Chemother. 2024 May 2;68(5):e0171923.

肥胖影響結核藥物的藥物動力學

參數	肥胖影響	臨床意義
分布體積 Vd	↑（脂溶性藥物如 RIF、PZA）	Cmax↓，AUC 下降，效果不足
清除率 CL	↑ 或變異度大	導致血中濃度下降，需要調整劑量或更密切監測
AUC	通常下降	AUC/MIC 不達標 → 治療失敗或延長療程

藥物	屬性	推薦劑量基準
Rifampin	脂溶性高	TBW 或 ABW
INH	中性	IBW 或 ABW
Pyrazinamide	親水性	IBW 或 ABW
Ethambutol	親水性	IBW 或 ABW

親水性藥物分布體積改變不大，但若用 TBW 劑量，容易過量。

Sandford Guide Antimicrobial

Drug	Recommendation	Comments
Aminoglycosides	ABW	Pharmacother 2007;27:1081. Follow levels so as to modify dose, if necessary, once hemodynamics stabilize.
Ethambutol	Ideal BW	Antimicrob Agents Chemother 2024;68:e0171923.
Isoniazid	Ideal BW	Antimicrob Agents Chemother 2024;68:e0171923.
Levofloxacin	No dose adjustment appears necessary	Data from 13 obese patients; variability in study findings renders conclusion uncertain. (Antimicrob Ag Chemother 55:3240, 2011; J Antimicrob Chemother 66:1653, 2011.) A PK study (requiring clinical validation) in patients with BMI ≥40 kg/m^2 suggests that higher doses may be necessary to achieve adequate drug exposure (Clin Pharmacokinet 53:753, 2014).
Moxifloxacin	No dose adjustment appears necessary	Antimicrob Agents Chemother2024;68:e0171923
Pyrazinamide	Ideal BW	Antimicrob Agents Chemother 2024;68:e0171923.
Rifampin	Insufficient data (TBW)	Current recommendation is for no dose adjustment (i.e., flat dosing), Limited data suggest a higher dose (with TDM) might be appropriate (Clin Pharmacokinet 2025;64:193; Antimicrob Agents Chemother 2024;68:e0171923).

臨床案例

38歲男性，92公斤，無職業，未婚，患有愛滋病，無其他慢性病史，咳嗽兩個月，診斷罹患活動性肺結核，痰塗及痰培養均為陽性，鑑定為MTB，藥敏試驗為全敏感；醫師給予標準抗結核藥物Rifater 5#QD，EMB 2#QD 。

Rifampicin Isoniazid and Pyrazinamide

1. 患者體重大於或等於50公斤者：5錠。
2. 患者體重介於40-49 公斤者：4 錠。
3. 患者體重介於30-39公斤者：3錠。

引用自113年結核病個案管理師教育訓練課程（複訓）課程

臨床案例

表 6-1 第一線抗結核藥物-單方藥物

藥物 (縮寫)	毒性/副作用	給藥方式	每日建議劑量依體重 (mg/Kg) (容許範圍) Max (最大劑量)	一般用法劑量	
				50Kg 以下	50Kg 以上
isoniazid (INH)	肝、神經、皮膚敏感	口服或肌肉/靜脈注射	5 (4-6) Max: 300 mg	100 mg/tab 2-3 顆	100 mg/tab 3 顆
rifampin (RMP)	體液/尿液變橘、肝、血液、胃腸不適、皮膚敏感	口服或靜脈注射	10 (8-12) Max: 600 mg	150 mg/tab 3 顆	300 mg/tab 2 顆
rifabutin (RFB)	肝、白血球低下、皮膚敏感、眼葡萄膜炎(uveitis)	口服	5 Max:300mg	≦30 kg: 150 mg 1 顆 31~45 kg: 150 mg 1.5 顆 ≧46 kg: 150 mg 2 顆	
pyrazinamide (PZA)	肝、高尿酸血症	口服	25 (15-30) Max: 2000 mg	40~55 kg: 500 mg/tab 2 顆 56~75 kg: 500 mg/tab 3 顆 ≧76 kg: 500 mg/tab 4 顆	
ethambutol (EMB)	視神經炎	口服	15 (15-20) Max:1600 mg	40~55 kg: 400 mg/tab 2 顆 56~75 kg: 400 mg/tab 3 顆 ≧76 kg: 400 mg/tab 4 顆	
streptomycin (SM) *	耳毒性、腎毒性、暈眩或聽力障礙	肌肉/靜脈注射	15(15-20) Max:1000 mg	500~750 mg	750 mg ~1000 mg

* 累積總劑量不超過 120 gm

EMB成人口服每日劑量15 (15-20) mg/kg-→ 92*15=1380/400= 3.45 tab

若有身高175公分

臨床案例

身體質量指數 (BMI) 計算器

世界衛生組織建議以身體質量指數 (Body Mass Index, BMI) 來衡量肥胖程度，其計算公式是以體重 (公斤) 除以身高 (公尺) 的平方。

身高 (公分)

請輸入身高

體重 (公斤)

請輸入體重

立即計算 全部清除

測量紀錄

總計測量 1 次，平均 BMI 為 30.0

身高 175 公分 體重 92 公斤 BMI 30 中度肥胖

Ideal Body Weight and Adjusted Body Weight

Calculates ideal body weight (Devine formula) and adjusted body weight.

INSTRUCTIONS

To calculate tidal volume by ideal body weight, use the ETT Depth and Tidal Volume Calculator.

When to Use | Pearls/Pitfalls | Why Use

Sex: Female **Male**

Height: 175 cm

Actual body weight: 92 lbs
Optional, for calculating adjusted body weight in obese patients

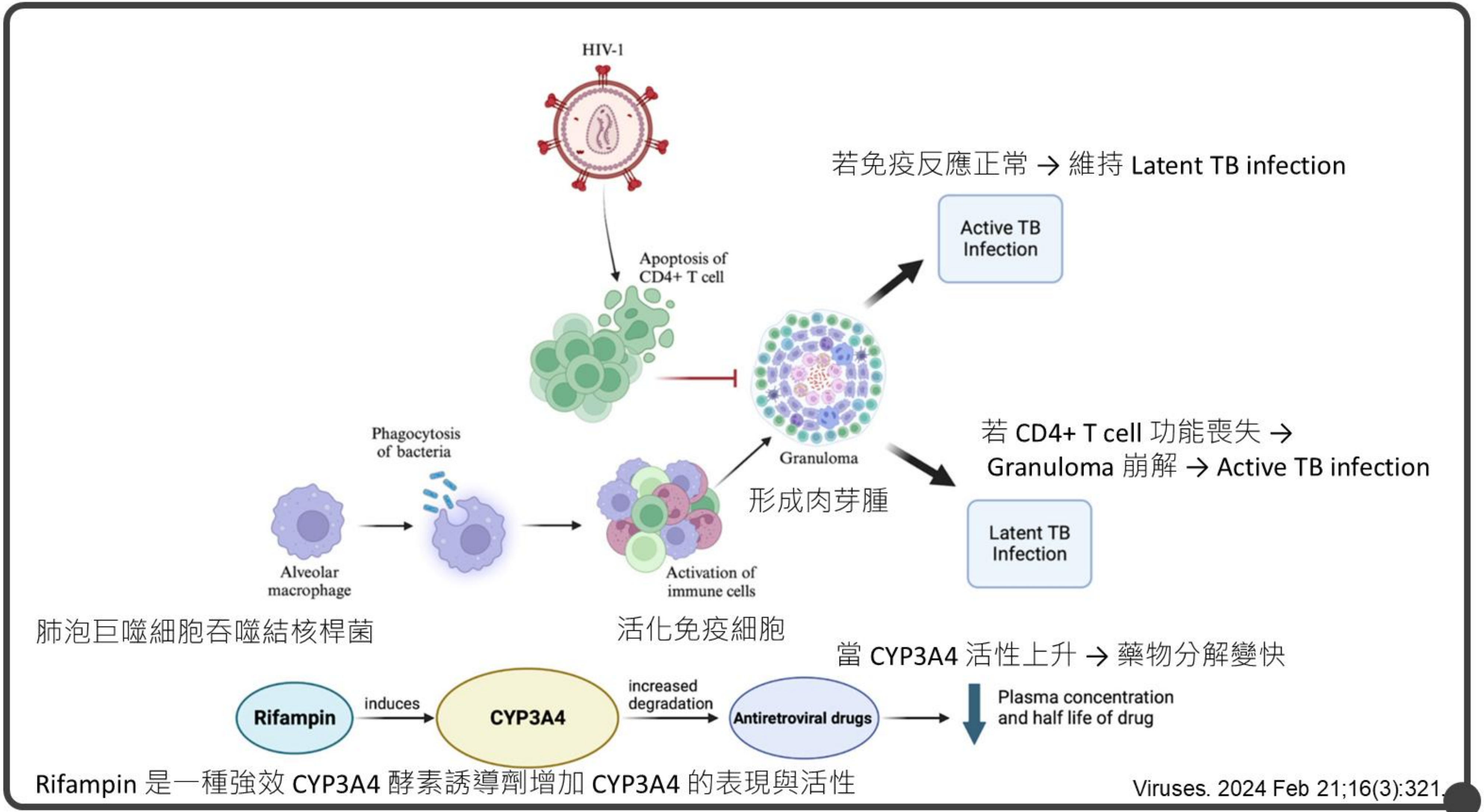
70 kg
Ideal Body Weight

Equivalent to 155 lbs

Actual body weight is 60% (0.6x) ideal body weight

Adjusted body weight only applicable if actual body weight is greater than ideal body weight

Copy Results | Next Steps »


HIV-1
Apoptosis of CD4+ T cell
若免疫反應正常 → 維持 Latent TB infection
Active TB Infection
Granuloma
形成肉芽腫
若 CD4+ T cell 功能喪失 → Granuloma 崩解 → Active TB infection
Latent TB Infection
Phagocytosis of bacteria
Alveolar macrophage
Activation of immune cells
肺泡巨噬細胞吞噬結核桿菌
活化免疫細胞
當 CYP3A4 活性上升 → 藥物分解變快
Rifampin
induces
CYP3A4
increased degradation
Antiretroviral drugs
Plasma concentration and half life of drug
Rifampin 是一種強效 CYP3A4 酵素誘導劑增加 CYP3A4 的表現與活性
Viruses. 2024 Feb 21;16(3):321.

HIV Drug Interactions

LIVERPOOL

TherEx

Centre for Experimental Therapeutics

Interaction Checker →

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Clinical Cases DDIs would like to hear from anyone who prescribed unboosted INSTIs + enzyme inducers. Click **here** to report your case

Interaction Checker

Access our free, comprehensive and user-friendly drug interaction charts

English

Español

Traducciones proporcionadas por Fundación Huésped

Français

Traductions fournies par Actions Traitements

Português

Traduções fornecidas pela Fundação Huésped

	Anti-tuberculosis drugs	ATV/c	ATV/r	DRV/c	DRV/r	LPV/r	DOR	EFV	ETV	NVP	RPV	FTR	MVC	LEN	BIC	CAB/RPV	DTG	EVG/c	RAL	TAF	TDF
First line and second line drugs	amikacin	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔ a
	bedaquiline	↑ b	↑ b	↑	↑	↑62% b	↔	↓18% ♥	↓	↑3%	↔ b	↔ b	↔	↑ c	↔	↔ b	↔	↑	↔	↔	↔
	capreomycin	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↑ d	↔	↔	↔	↑ E a
	clofazimine	↔	↔	↔	↔	↔	E	↔ ♥	↔	↔	E	E	E	↔	E	E	↔	↔	↔	↔	↔
	cycloserine	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔
	delamanid	e	e	e	e	e	↔	↔ f ♥	↔	↔	↔ g	↔ g	↔	h	↔	↔ g	↔	e	↔	↔	↔
	ethambutol	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔
	ethionamide	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔
	isoniazid	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔
	kanamycin	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔ a
	linezolid	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔
	moxifloxacin	↑ b	↓ b	↔	↓	↓ b	↔	↓ ♥	↓	↔	↔ b	↔ b	↔	↔	↔	↔ b	↔	↔	↔	↔	↔
	para-aminosalicylic acid	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↑	↔	↔	↔	↑E
	pretomanid	↓ b	↓ b	↓	↓	↓17% b	↔	↓35% ♥	↓	↓	↔ b	↔ b	↔	↔	↔	↔ b	↔	↓	↔	↔	↔
	pyrazinamide	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔
	rifabutin	↑ D i	↑ j	↑ D i	↑ j	↑ j	D50% k	↓38% l	D37%	↑17%	D42% m	D30%	n	D #	D38%	D	↔	↑ D i	E19%	D o	↔
	rifampicin	D	D72%	D	D57%	D75% p	D82%	D26% q	D	D58%	D80%	D82%	D r	D82% #	D75%	D	D54% s	D	D40% t	D o	D12%
	rifapentine	D	D	D	D	D	D	D	D	D	D	D	D r	D #	D	D	D u	D	D	D o	↔
	streptomycin	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔	↔ a

EACS Guidelines – Drug–Drug Interactions: Anti-Tuberculosis Drugs and ARVs

Anti-TB Drugs	ARV Agents	PK Interaction	Clinical Comments and Recommended Doses
Rifampin	All PIs, DOR, ETR, NVP, RPV (IM and PO), CAB (IM and PO), BIC and EVG/c	– ARVs concentrations ↓ > 50%.	– Not recommended
	EFV	– AUC of EFV ↓ 26%	– EFV 600 mg once daily (but not 400 mg once daily) is recommended to use with rifampin; monitoring virologic response is needed.
	TAF	– Compared with TDF alone, TAF plus rifampin, AUC of TFV-DP ↑ 4.2-fold; – Compared with TAF alone, TAF plus rifampin, AUC of TAF ↓ 55%, AUC of TFV-DP ↓ 36%; – Compared with TAF 25 mg once daily alone, TAF 25 mg twice daily plus rifampin, AUC of TAF ↓ 14%, AUC of TFV-DP ↓ 24%.	– Avoid concurrent administration unless benefits outweigh risks. If used concomitantly, closely monitor antiviral efficacy.
	DTG	– Compared to DTG 50 mg twice daily, rifampin with DTG 50 mg twice daily, AUC of DTG ↓ 54%, C_{min} ↓ 72%; – Compared to DTG 50 mg once daily, rifampin with DTG 50 mg twice daily, AUC of DTG ↑ 33% and C_{min} ↑ 22%.	– DTG 50 mg twice daily for patients is recommended if without suspected or confirmed INSTI-related resistance mutations. Use rifabutin instead of rifampin if with INSTI-related mutations.
	RAL	– Compared to RAL 400 mg twice daily, RAL 400 mg per day, AUC of RAL ↓ 40% and C_{min} ↓ 61%; – Compared to RAL 400 mg twice daily, rifampin with RAL 800 mg twice daily, AUC of RAL ↑ 27% and C_{min} ↓ 53%.	– RAL 800 mg twice daily is recommended with monitoring antiviral efficacy closely or replace rifampin with rifabutin. RAL 1200 mg once daily plus rifampin is not recommended.
	MVC *	– AUC of MVC ↓ 63%.	– Increase MVC dose to 600 mg twice daily.
	TDF	– No significant influence	– No dose adjustment needed

Viruses. 2024 Mar 23;16(4):494.

Anti-TB Drugs	ARV Agents	PK Interaction	Clinical Comments and Recommended Doses
Rifabutin	PIs/c, RPV(IM), TAF, BIC, CAB (IM), and EVG/c	– Boold concentrations of those ARV agents decrease significantly	– Not recommended
	PIs/r	– Concentrations of rifabutin or its metabolite would increase significantly	
	DOR	– AUC of DOR ↓ 50%	– Increase DOR dose to 100 mg twice per day. There is no need to adjust rifabutin dose.
	EFV	– Rifabutin ↓ 38%	– The suggested rifabutin dose range is 450–600 mg per day.
	RPV (PO)	– Compared to RPV 25 mg once daily, the co-administration of RPV 50 mg once daily with rifabutin, AUC and C_{min} of RPV ↔	– The RPV dose is increased to 50 mg once daily. No dose adjustment is needed for rifabutin.
	ETR	– AUC of rifabutin and metabolite ↔; AUC of ETR ↓ 37%	– Rifabutin is not suggested to co-administer with ETR plus PIs/r. Without PIs/r, rifabutin 300 mg once daily plus ETR is suitable.
	MVC	– AUC of MVC ↔ and C_{min} ↓ 30%	– MCV 150 mg twice daily is recommended when used with a strong CYP3A inhibitor, while 300 mg twice daily is suggested if not.
	NVP, TDF, DTG, RAL, and CAB (PO)	– No significant influence	– No dose adjustment needed
Rifapentine	All PIs, DOR, ETR, NVP, RPV (IM and PO), TAF, BIC, EVG/c, CAB (IM and PO), and MVC	– Concentrations of those ARVs decrease significantly	– Not recommended
	DTG	– AUC of DTG ↓ 26% and C_{min} ↓ 47% when co-administrated with rifapentine 900 mg once weekly	– DTG 50 mg per day may be suggested to co-administer with once weekly rifapentine (but not once daily rifapentine) in individuals with viral suppression, and it is needed to monitor antiviral efficacy.
	RAL	– AUC of RAL ↑ 71% and C_{min} ↓ 12% when co-administrated with rifapentine 900 mg once weekly; – C_{min} of RAL ↓ 41% when co-administrated with rifapentine 600 mg once daily.	– RAL 400 mg twice daily is recommended with once weekly rifapentine (but not once daily rifapentine) without dose adjustment.
	EFV and TDF	– No significant influence	– No dose adjustment needed

Viruses. 2024 Mar 23;16(4):494.

表 9-1 國內現有常用抗愛滋病毒藥物與 rifamycin 類之交互作用

Antiretroviral agent	Rifampin	Rifabutin	Rifapentine
Non-nucleoside reverse transcription inhibitors （NNRTIs）			
Efavirenz (EFV)	EFV 濃度↓26%，可維持正常劑量。Rifampin 是使用 EFV 時首選的抗結核藥物。	Rifabutin ↓ 38%，併用會降低 rifabutin 濃度，建議 rifabutin 劑量為 450 mg QD。	EFV 600 mg 可維持正常劑量。
Etravirine (ETR)	可能大幅降低 ETR 濃度，不建議和 rifampin 併用。	ETR AUC↓37%，rifabutin 和 rifabutin 代謝物 AUC↓17%，可維持 rifabutin 劑量 300mg QD。若 ETR 合併使用 darunavir/ritonavir，不應選用 rifabutin 治療結核病。	可能大幅降低 ETR 濃度，不建議同時使用。
Rilpivirine (RPV)	RPV 濃度↓80% & AUC↓80%，兩者不能併用。	RPV 濃度↓31% & AUC↓42%，RPV 劑量應由 25 mg QD 調整至 50mg QD。	RPV 濃度↓，兩者不能併用。
Doravirine (DOR)	DOR AUC↓88%，兩者不能併用。	DOR AUC ↓ 50%，建議 DOR 由 100 mg QD 增加劑量到 100 mg 每日兩次使用。Rifabutin 不需改變劑量	DOR 濃度↓，兩者不能併用。

Protease inhibitors （PIs）			
Atazanavir (ATV)	PI 濃度↓大約 75%；所有蛋白酶抑制劑都不建議和 rifampin 併用。 Rifampin	ATV 濃度沒有影響，但是會增加 rifabutin 濃度。建議 rifabutin 劑量由原來 300mg QD 減為 150 mg QD。 Rifabutin	Rifapentine 對於 CYP3A4 induction 的效果雖然比 rifampin 差，但比 rifabutin 強。同時使用預計會下降 ATV 濃度，因此不建議。 Rifapentin
Lopinavir/ritonavir（LPV/r）	PI 濃度↓大約 75%；所有蛋白酶抑制劑都不建議和 rifampin 併用。	併 LPV/r，和單用 rifabutin 300 mg 相比，rifabutinAUC ↑473%。因此，rifabutin 劑量為 150 mg QD。	Rifapentine 對於 CYP3A4 induction 的效果雖然比 rifampin 差，但比 rifabutin 強。同時使用預計會下降 LPV/r 濃度，因此不建議。
Darunavir/cobicistat（DRV/c）	PI 濃度↓大約 75%；所有蛋白酶抑制劑都不建議和 rifampin 併用。	Rifabutin 濃度增加，DRV/c 濃度下降，不建議同時使用。	Rifapentine 對於 CYP3A4 induction 的效果雖然比 rifampin 差，但比 rifabutin 強。同時使用預計會下降 DRV/c 濃度，因此不建議。

Antiretroviral agent	Rifampin	Rifabutin	Rifapentine
Integrase strand transfer inhibitors （INSTIs）			
Raltegravir (RAL)	RAL 400 mg BID 併用 rifampin 時，AUC↓40%；濃度↓61%。併用時需把 RAL 建議劑量由 400 mg BID 改為 800 mg BID。	RAL 400 mg BID 併用 rifabutin 時，AUC↑19%，不需調整劑量。	• RAL 400 mg BID 併用 rifapentine 900 mg once weekly 使用時，RAL↑71%，不需調整劑量。 • 但是如果併用 rifapentine 600 mg QD 使用，RAL Cmin ↓41%，因此不建議與 rifapetnine 600 mgQD 同時使用。
Elvitegravir/cobicistat (EVG/c)	不能併用，因 Elvitegravir 及 cobicistat 皆為 CYP3A4 受質，併用 rifampin 可能嚴重降低病毒抑制效果而促使抗藥性的產生。	EVG/c 併用 rifabutin (150 mg 隔日服用一次)與單獨使用 rifabutin 300 mg QD 相比，rifabutin 濃度相當，但是 EVG 的 AUC↓21%,Cmin↓67%。因此不建議同時使用。	EVG/c 併用 rifapentine 會降低 cobicistat 濃度繼而減少 EVG 濃度，因此不建議同時使用。

Antiretroviral agent	Rifampin	Rifabutin	Rifapentine
Integrase strand transfer inhibitors （INSTIs）			
Dolutegravir (DTG)	DTG 50 mg BID 合併併用 rifampin 時，AUC 比起 DTG 50mg BID 單獨使用↓40%，比起 DTG 50 mg QD 單獨使用↑33%。因此建議併用時需把 DTG 建議劑量由 50 mg QD 改為 50 mg BID。	DTG 50 mg QD 併用 rifabutin 300 mg QD 時，DTG 的 AUC 與單獨使用時相當，因此不需調整劑量。	rifapentine 900 mg + isoniazid once weekly (3HP) 與 DTG 50 mg QD 合併使用，會減少 29% DTG 濃度，不過病人血漿中愛滋病毒量依舊可以維持檢測不到。1HP 和 DTG 併用目前尚沒有資料。
Bictegravir (BIC)	併用 rifampin 時，BIC AUC ↓75%，因此不建議同時使用。	BIC 併用 rifabutin300 mg QD 時，BIC AUC↓38%，因此不建議同時使用。	根據臺大醫院針對 48 位併用 BIC/FTC/TAF 與 1HP 的藥物濃度與病毒量控制的研究結果，rifapentine 會顯著降低 BIC 濃度。但是對於 1HP 治療結束後，病毒量控制並沒有受到影響，建議謹慎追蹤病毒量變化。

Rifabutin–Cobicistat Drug Interaction

- 43 歲女性
- HIV/AIDS 併發肺結核
- ART regimen ： Dolutegravir + Darunavir/Cobicistat + Emtricitabine/ Tenofovir disoproxil fumarate
- 抗結核治療：Rifampin，但因與 PI交互作用，改用 Rifabutin

Case Rep Ophthalmol 2020;11:156–160

Do Not Coadminister
Darunavir/cobicistat (DRV/c)
Rifampicin

Quality of evidence: Very Low

Summary:

Coadministration has not been studied and is contraindicated as it may significantly decrease darunavir/cobicistat concentrations, leading to loss of therapeutic effect and possible development of resistance.

Description:

Based on theoretical considerations rifampin is expected to decrease darunavir and/or cobicistat plasma concentrations (CYP3A induction). The combination of rifampicin and darunavir/cobicistat is contraindicated due to the potential for loss of therapeutic effect.

Rezolsta Summary of Product Characteristics, Janssen-Cilag Ltd, July 2023.

Coadministration is expected to decrease concentrations of darunavir and cobicistat. Coadministration is contraindicated with rifampicin due to the potential for reduced plasma concentrations of darunavir, which may result ir loss of therapeutic effect and development of resistance.

Prezcobix US Prescribing Information, Janssen Pharmaceuticals Inc, March 2023.

Potential Interaction
Darunavir/cobicistat (DRV/c)
Rifabutin

Quality of evidence: Very Low

Summary:

Coadministration of darunavir/cobicistat and rifabutin has not been studied and is not recommended as it may significantly decrease cobicistat plasma concentrations and consequently those of darunavir being boosted. Coadministration of elvitegravir/cobicistat (150 mg/150 mg once daily) and rifabutin (150 mg every other day) decreased cobicistat Ctrough by 66%. Rifabutin exposure was similar to values obtained alone, but 25-O-desacetylrifabutin exposures were increased (AUC, Cmax and Cmin were increased by 525%, 384% and 394%). The European SPC for darunavir/cobicistat recommends that if the combination is needed, to use rifabutin 150 mg 3 times per week on set days (e.g. Monday-Wednesday-Friday); the US prescribing information for darunavir/cobicistat recommends rifabutin 150 mg every other day. Both labels recommend increased monitoring for rifabutin associated adverse reactions including neutropenia and uveitis due to increased rifabutin exposure. Further dose reduction of rifabutin has not been studied and a twice weekly dose of 150 mg may not provide an optimal exposure to rifabutin leading to a risk of rifamycin resistance and a treatment failure.

Title Rifabutin / Cobicistat

Risk Rating D: Consider therapy modification

Summary Cobicistat may increase serum concentration of Rifabutin. **Severity** Major **Reliability Rating** Intermediate

Patient Management Avoid coadministration of cobicistat and rifabutin if possible. Clinical practice guidelines state rifabutin and cobicistat should not be coadministered. Cobicistat labeling recommends decreasing the rifabutin dose to 150 mg every other day. If combined, monitor patients for increased rifabutin toxicities (eg, neutropenia, uveitis).

Discussion Rifabutin concentrations were not significantly altered with concurrent use of elvitegravir/cobicistat in a study summarized in US prescribing information.[1,2] However, concentrations of the active metabolite 25-O-desacetyl-rifabutin were significantly increased, with the AUC 6.25-fold higher with concurrent use of lower-dose rifabutin (150 mg every other day) and elvitegravir/cobicistat (150 mg/150 mg once daily). No studies have evaluated the effects of cobicistat-boosted protease inhibitor therapy on rifabutin concentrations. However, ritonavir-boosted protease inhibitor regimens have been shown to increase the AUC of rifabutin and its active metabolite substantially, and similar effects are expected with cobicistat.[3]

Cobicistat labeling recommends decreasing the rifabutin dose to 150 mg every other day.[3] Due to the expected increase in rifabutin concentrations, clinical practice guidelines do not recommend use of rifabutin in patients treated with cobicistat.[4] Patients should be monitored for increased rifabutin effects/toxicities (eg, neutropenia, uveitis) if these agents are combined.[3]

The specific mechanism for this interaction is uncertain but appears to be at least in part due to inhibition of CYP3A-mediated rifabutin metabolism by cobicistat.

● Do Not Coadminister ■ Potential Interaction ▲ Potential Weak Interaction ◆ No Interaction Expected

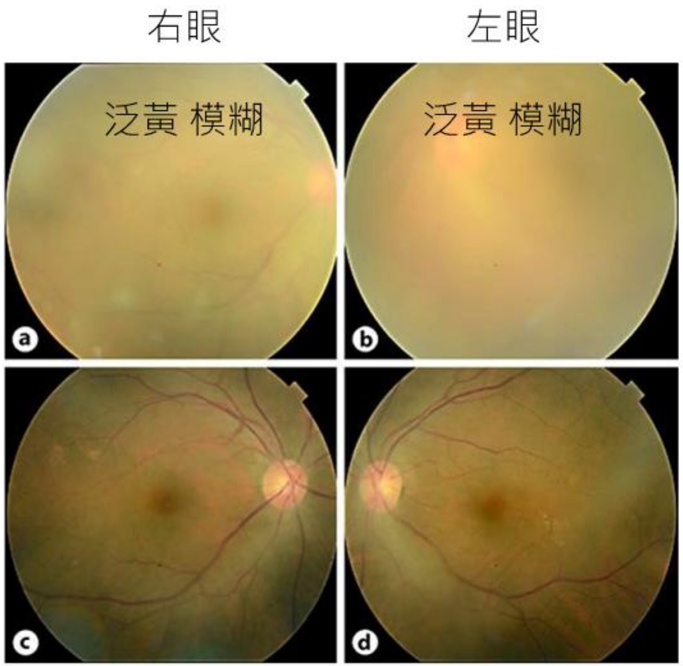
Results Key

	DRV/c	DTG	FTC	TDF				
Rifabutin	■	◆	◆	◆	■	◆	◆	◆
Rifampicin	●	■	◆	◆	●	■	◆	◆
Rifapentine	●	■	◆	◆	●	■	◆	◆

- 服用 Rifabutin + Cobicistat 第 14 天
嚴重視力模糊、眼球疼痛、光敏感與畏光

- 眼科檢查
雙眼葡萄膜炎（anterior uveitis）
玻璃體混濁與視網膜炎（vitritis + retinitis）

- 藥物血中濃度監測
Rifabutin 416 ng/ml (Reference 50-150 ng/mL)



玻璃體清晰、黃斑與optic disc清楚可見

Case Rep Ophthalmol 2020;11:156–160

臨床案例

Dovato(Dolutegravir/Lamivudine)或
Triumeq(Dolutegravir/Abacavir/Lamivudine和RIF併用時
臨床上如何處理?

Antiretroviral agent	Rifampin	Rifabutin	Rifapentine
Integrase strand transfer inhibitors （INSTIs）			
Dolutegravir (DTG) 增加DTG劑量!	DTG 50 mg BID 合併併用 rifampin 時，AUC 比起 DTG 50mg BID 單獨使用↓40%，比起 DTG 50 mg QD 單獨使用↑33%。因此建議併用時需把 DTG 建議劑量由 50 mg QD 改為 50 mg BID。	DTG 50 mg QD 併用 rifabutin 300 mg QD 時，DTG 的 AUC 與單獨使用時相當，因此不需調整劑量。	rifapentine 900 mg + isoniazid once weekly (3HP) 與 DTG 50 mg QD 合併使用，會減少 29% DTG 濃度，不過病人血漿中愛滋病毒量依舊可以維持檢測不到。1HP 和 DTG 併用目前尚沒有資料。
Bictegravir (BIC)	併用 rifampin 時，BIC AUC↓75%，因此不建議同時使用。	BIC 併用 rifabutin300 mg QD 時，BIC AUC↓38%，因此不建議同時使用。	根據臺大醫院針對 48 位併用 BIC/FTC/TAF 與 1HP 的藥物濃度與病毒量控制的研究結果，rifapentine 會顯著降低 BIC 濃度。但是對於 1HP 治療結束後，病毒量控制並沒有受到影響，建議謹慎追蹤病毒量變化。

Cape Town
South Africa

Standard-dose versus double-dose dolutegravir in HIV-associated tuberculosis in South Africa (RADIANT-TB): a phase 2, non-comparative, randomised controlled trial

Rulan Griesel, Ying Zhao, Bryony Simmons, Zaayid Omar, Lubbe Wiesner, Claire M Keene, Andrew M Hill, Graeme Meintjes, Gary Maartens

資源匱乏地區有時候很難做到

Summary

Background The drug–drug interaction between rifampicin and dolutegravir can be overcome by supplemental dolutegravir dosing, which is difficult to implement in high-burden settings. We aimed to test whether virological outcomes with standard-dose dolutegravir-based antiretroviral therapy (ART) are acceptable in people with HIV on rifampicin-based antituberculosis therapy.

Methods RADIANT-TB was a phase 2b, randomised, double-blind, non-comparative, placebo-controlled trial at a single site in Khayelitsha, Cape Town, South Africa. Participants were older than 18 years of age, with plasma HIV-1 RNA greater than 1000 copies per mL, CD4 count greater than 100 cells per µL, ART-naive or first-line ART interrupted, and on rifampicin-based antituberculosis therapy for less than 3 months. By use of permuted block (block size of 6) randomisation, participants were assigned (1:1) to receive either tenofovir disoproxil fumarate, lamivudine, and dolutegravir plus supplemental 50 mg dolutegravir 12 h later or tenofovir disoproxil fumarate, lamivudine, and dolutegravir plus matched placebo 12 h later. Participants received standard antituberculosis therapy (rifampicin, isoniazid, pyrazinamide, and ethambutol for the first 2 months followed by isoniazid and rifampicin for 4 months). The primary outcome was the proportion of participants with virological suppression (HIV-1 RNA <50 copies per mL) at week 24 analysed in the modified intention-to-treat population. This study is registered with ClinicalTrials.gov, NCT03851588.

Findings Between Nov 28, 2019, and July 23, 2021, 108 participants (38 female, median age 35 years [IQR 31–40]) were randomly assigned to supplemental dolutegravir (n=53) or placebo (n=55). Median baseline CD4 count was 188 cells per µL (IQR 145–316) and median HIV-1 RNA was 5·2 $\log_{10}$ copies per mL (4·6–5·7). At week 24, 43 (83%, 95% CI 70–92) of 52 participants in the supplemental dolutegravir arm and 44 (83%, 95% CI 70–92) of 53 participants in the placebo arm had virological suppression. No treatment-emergent dolutegravir resistance mutations were detected up to week 48 in the 19 participants with study-defined virological failure. Grade 3 and 4 adverse events were similarly distributed between the study arms. The most frequent grade 3 and 4 adverse events were weight loss (4/108 [4%]), insomnia (3/108 [3%]), and pneumonia (3/108 [3%]).

Interpretation Our findings suggest that twice-daily dolutegravir might be unnecessary in people with HIV-associated tuberculosis.

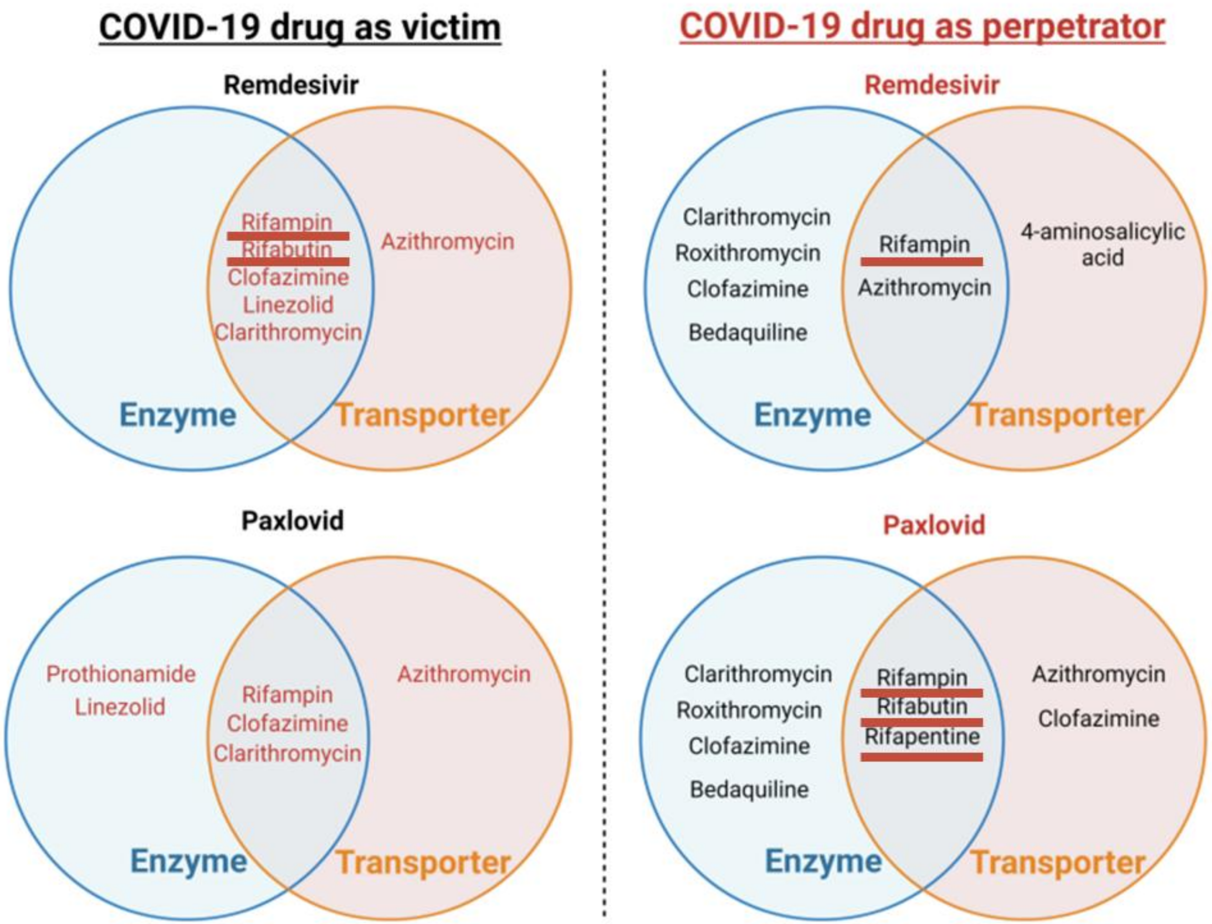
Lancet HIV 2023; 10: e433–41

Published Online May 22, 2023 https://doi.org/10.1016/S2352-3018(23)00081-4

See Comment page e422

Division of Clinical Pharmacology, Department of Medicine, University of Cape Town, Cape Town, South Africa (R Griesel MMed, L Wiesner PhD, Prof G Maartens MMed); Wellcome Centre for Infectious Diseases Research in Africa, Institute of Infectious Disease and Molecular Medicine, and Department of Medicine, University of Cape Town, Cape Town, South Africa (R Griesel, Y Zhao MMed, Z Omar MBChB, Prof G Meintjes PhD, Prof G Maartens); Department of Medicine, University of Cape Town, Cape Town, South Africa (Y Zhao, Prof G Meintjes); LSE Health, London School of Economics and Political Science, London, UK (B Simmons PhD); Médecins Sans Frontières, Cape Town, South Africa (C M Keene MSc); Centre for Tropical Medicine

Covid-19
藥物可能是被害者
也可能是加害者



Pharmaceuticals (Basel). 2024 Aug 6;17(8):1035.

COVID-19 Drug Interactions

UNIVERSITY OF LIVERPOOL

TherEx
Centre for Experimental Therapeutics

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Follow us on Bluesky @covidinteractions for interaction news and for the latest additions and changes to the website

Interaction Checker

Access our free, comprehensive and user-friendly drug interaction charts

Discover Our COVID-19 iChart Mobile App

COVID-19 iChart gives easy access to our drug interaction information on mobile devices.
Click the links below to get the app for your iPhone or Android device.

Download on the App Store

GET IT ON Google Play

● Do Not Coadminister ■ Potential Interaction △ Potential Weak Interaction ◆ No Interaction Expected

Results Key

	Ensitrelvir	Molnupiravir	Nirmatrelvir/ritonavir (5 days)	Remdesivir
Bedaquiline	■	◆	■	△
Cycloserine	◆	◆	◆	◆
Delamanid	■	◆	■	△
Ethambutol	◆	◆	◆	◆
Ethionamide	◆	◆	◆	◆
Isoniazid	◆	◆	◆	◆
Levofloxacin	◆	◆	◆	■
Moxifloxacin	◆	◆	◆	■
Para-aminosalicylic acid	◆	◆	◆	◆
Pyrazinamide	◆	◆	◆	◆
Rifabutin	●	◆	■	◆
Rifampicin (Rifampin)	●	◆	●	△
Rifapentine	●	◆	●	△
Streptomycin	◆	◆	◆	◆

Do Not Coadminister

Nirmatrelvir/ritonavir (5 days)

Rifampicin (Rifampin)

Quality of Evidence: Very Low

Summary:

Coadministration of nirmatrelvir/ritonavir with rifampicin is contraindicated as it may cause large decreases in nirmatrelvir/ritonavir concentrations which may in turn significantly decrease the nirmatrelvir/ritonavir therapeutic effect. Due to the persisting inducing effect upon discontinuation of a strong inducer, consider an alternative COVID-19 treatment.

Description:

Coadministration is contraindicated. Decreased plasma concentrations of nirmatrelvir/ritonavir may lead to loss of virologic response and possible resistance. Rifampicin is strong CYP3A4 inducer, and this may lead to a decreased exposure of nirmatrelvir/ritonavir and potential loss of virologic response.

Paxlovid GB Summary of Product Characteristics, Pfizer Ltd, December 2024.

台灣感染症醫學會
The Infectious Diseases Society of Taiwan

第三版　版本日期：2023 年 12 月 20 日

Paxlovid™（倍拉維）與常見用藥交互作用管理建議

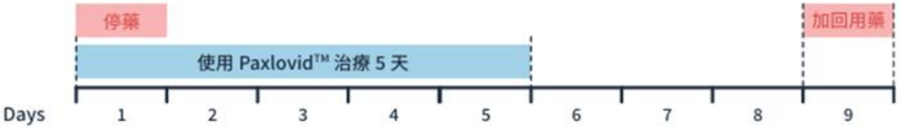
Paxlovid™（Nirmatrelvir + ritonavir）

適用於 COVID-19 發病五天內、未使用氧氣且具重症危險因子的輕度至中度確診患者，患者須滿 12 歲體重在 40 公斤以上[1,2]。三期臨床試驗結果顯示對此族群患者可減少 89% 的死亡或住院風險[3]。

用法用量

口服 Nirmatrelvir 300 mg + ritonavir 100 mg（即 2#/1#），3 顆錠劑一同服用，每日兩次，持續 5 天[1,2]。

若患者目前使用以下藥物，建議停藥或使用替代藥物，並使用 Paxlovid。
無法停藥或使用其他替代用藥時，建議改用其他 COVID-19 抗病毒藥物[2,4-7]



抗心律不整	抗凝血劑	抗血小板	降血脂 &	心衰竭	肺高壓
Quinidine Amiodarone (CORDARONE®)¹ Dronedarone (Multaq®) Propafenone (RYTMONORM®) Flecainide (TAMBOCOR®)	Rivaroxaban (Xalrelto®)	Ticagrelor (BRILINTA®) Clopidogrel (PLAVIX®)*	Lovastatin (MEVACOR®) Simvastatin (ZOCOR®) Atorvastatin (LIPITOR®) Rosuvastatin (CRESTOR®)	Eplerenone (Inspra®) Ivabradine (Coralan®)	Tadalafil (Cialis®) Vardenafil (Levitra®) Sildenafil (VIAGRA®, REVATIO®)
抗癲癇藥	**偏頭痛**	**鎮靜安眠**	**抗精神病**	**慢性腎臟病**	**抗微生物**
Carbamazepine (Tegretol®)' Phenytoin (DILANTIN®)' Phenobarbital' Primidone (PRIMACLONE®)'	Ergotamine (DIHYDERGOT®) Eletriptan (Eletriptan®)	Midazolam (DORMICUM®) Diazepam	Lurasidone (Latuda®) Quetiapine (SEROQUEL®)	Finerenone (Kerendia®)	Rifampin (RIFAMPICIN®)' Glecaprevir/pibrentasvir (MAVIRET®) Rifapentine (Priftin®)

性別	男	年齡	53.9歲	**Ht/Wt**	169cm/45.6kg
住家電話	---			行動	---

Bilirubin-T: 0.5(25/04/03)	**Creatinine:** 0.60(25/04/03)	**CrCl:** 155.54mL/min	**Estimated GFR:** 140.94(25/04/03)
GOT (AST): 31(25/04/03)	**GPT (ALT):** 14(25/04/03)	**Bilirubin-T:** 0.4(24/12/26)	**GPT (ALT):** 13(24/12/26)
GOT (AST): 37(24/12/26)	**Estimated GFR:** 130.82(24/12/26)	**Creatinine:** 0.64(24/12/26)	**CrCl:** 145.82mL/min
Bilirubin-T: 0.8(24/09/23)	**Creatinine:** 0.62(24/09/23)	**CrCl:** 150.52mL/min	**Estimated GFR:** 135.70(24/09/23)
GOT (AST): 26(24/09/23)	**GPT (ALT):** 9(24/09/23)	**Creatinine:** 0.66(24/05/07)	**CrCl:** 141.4mL/min
GPT (ALT): 17(24/05/07)	**GOT (AST):** 38(24/05/07)	**Estimated GFR:** 126.75(24/05/07)	

科別/醫師/日期/處方別/Indication	註	藥名	用量	用法	天數	總量
感染科- 2(B20 Human immunodeficiency virus [HIV] disease U07.1 COVID-19		**1**.Nirmatrelvir/Ritonavir Tab.(疾管署) 150mg/100mg/Tab.(Paxlovid Tab.(疾管署))	3tab	Q12H-PO	5	1**box**
感染科- 2(A31.0 Pulmonary mycobacterial infection B18.1 Chronic viral hepatitis B without delta-agent B20 Human immunodeficiency virus [HIV] disease B37.7 Candidal sepsis F20.0 Paranoid schizophrenia J18.9 Pneumonia, unspecified organism L28.0 Lichen simplex chronicus		**1**.Rifabutin Cap. 150mg/cap(Mycobutin Cap.)	2cap	QD-PO	30	60**cap**
		2.Darunavir/cobicistat Tab.(急採) 800mg/150mg(Prezcobix Tab.(急採))	1tab	QD-PO	30	30**tab**
		3.【錠】Levofloxacin TAB. 500mg/tab(Cravit TAB.)	1tab	QD-PO	30	30**tab**
		4.Ethambutol 400mg/ tab(E-BUTOL)	2tab	QD-PO	30	60**tab**
		5.Dextromethorphan TAB. 30mg/tab(Mephan TAB.)	1tab	TID-PO	30	90**tab**
		6.【錠】Clarithromycin TAB. 500mg/tab(Klaricid TAB.)	1tab	Q12H-PO	30	60**tab**
		7.Bictegravir/FTC/TAF Tab. 50mg/200mg/25mg/tab(Biktarvy Tab.)	1tab	QD-PO	30	30**tab**
		8.Sulfamethoxazole 400mg & Trimethoprim 80mg 400mg/80mg/tab(Morcasin TAB.)	1tab	QD-PO	30	30**tab**
		9.Acetylcysteine TAB. 600mg/tab(Fluimucil A Effervescent TAB.)	1tab	BID-SOLVE	30	60**tab**
		10.Nystatin SUSP. 2.4MIU/24 mL(Mycostatin SUSP.)	2mL	Q6H-PO	7	3**btl**

< Back To Checker

RIFABUTIN

MAJOR

Warning:

Concurrent use of NIRMATRELVIR/RITONAVIR and RIFABUTIN may result in increased rifabutin exposure, increased rifabutin-related adverse effects, reduced nirmatrelvir/ritonavir exposure and reduced efficacy of nirmatrelvir/ritonavir.

Clinical Management:

Concomitant administration of rifabutin with nirmatrelvir/ritonavir may increase rifabutin exposure. When coadministration of rifabutin and nirmatrelvir/ritonavir is necessary, the rifabutin dose should be reduced by at least 75% of the usual dose (300 mg/day); not to exceed a maximum dose of 150 mg every other day or 3 times per week. Further dosage reduction of rifabutin may be warranted, and more frequent monitoring for rifabutin adverse effects is recommended. The risk of adverse events associated with rifabutin, including uveitis, may be increased (Pfizer Inc, 2021). Additionally, concomitant use of nirmatrelvir/ritonavir and rifabutin (CYP3A inducer) may reduce nirmatrelvir/ritonavir exposure and may lead to potential loss of virologic response and possible resistance (Prod Info PAXLOVID(TM) oral tablets, 2024).

Potential Interaction

Nirmatrelvir/ritonavir (5 days)

Rifabutin

Quality of Evidence: Very Low

Summary:

Coadministration has not been studied. Rifabutin is mainly metabolized by CYP3A4 and cholinesterase. Coadministration may increase exposure of rifabutin (due to inhibition of CYP3A4 by nirmatrelvir/ritonavir). Administering rifabutin 150 mg every other day with ritonavir boosted HIV protease inhibitors has been shown to result in sub-optimal rifabutin concentrations. Thus, it is recommended to give rifabutin 150 mg every day in presence of nirmatrelvir/ritonavir. After stopping nirmatrelvir/ritonavir, the CYP3A4 inhibitory effect of nirmatrelvir/ritonavir is predicted to mostly disappear after 3 days.

X	Avoid combination
D	Consider therapy modification
C	Monitor therapy
B	No action needed
A	No known interaction

More about Risk Ratings ▼

X	Tacrolimus (Systemic) CycloSPORINE (Systemic)
X	Tacrolimus (Systemic) Sirolimus (Oral) (Sirolimus (Conventional)) (Sirolimus Products)
D	Everolimus CycloSPORINE (Systemic)
D	Mycophenolate CycloSPORINE (Systemic)
D	RifAMPin Mycophenolate
D	RifAMPin (CYP3A4 Inducers (Strong)) CycloSPORINE (Systemic)
D	RifAMPin (CYP3A4 Inducers (Strong)) Sirolimus (Oral) (Sirolimus (Conventional))
D	RifAMPin (CYP3A4 Inducers (Strong)) Tacrolimus (Systemic)
D	RifAMPin (Inducers of CYP3A4 (Strong) and P-glycoprotein) Everolimus
D	Sirolimus (Oral) (Sirolimus (Conventional)) CycloSPORINE (Systemic)

C	Everolimus (CYP3A4 Inhibitors (Weak)) Sirolimus (Oral) (Sirolimus (Conventional))
C	Isoniazid (CYP3A4 Inhibitors (Weak)) CycloSPORINE (Systemic)
C	Isoniazid (CYP3A4 Inhibitors (Weak)) Sirolimus (Oral) (Sirolimus (Conventional))
C	Isoniazid (CYP3A4 Inhibitors (Weak)) Tacrolimus (Systemic)
C	Mycophenolate Ethambutol (Antibiotics)
C	Mycophenolate Isoniazid (Antibiotics)
C	Mycophenolate Pyrazinamide (Antibiotics)
C	Mycophenolate Rifabutin (Antibiotics)
C	Mycophenolate Rifapentine (Antibiotics)
C	Pyrazinamide CycloSPORINE (Systemic)

Title CYP3A4 Inducers (Strong) / Tacrolimus (Systemic)

Risk Rating D: Consider therapy modification

Summary CYP3A4 Inducers (Strong) may decrease serum concentration of Tacrolimus (Systemic). **Severity** Moderate **Reliability Rating** High-Intermediate

Patient Management Tacrolimus dose increases will likely be needed during concomitant use with strong CYP3A4 inducers. Monitor more closely and frequently for decreased tacrolimus concentrations and effects when combined. Maximal reductions in tacrolimus exposure are expected to occur after multiple doses of the strong CYP3A4 inducer have been administered, within days to weeks of initiation. Induction effects may also persist for days to weeks after the inducer has been discontinued.

CYP3A4 Inducers (Strong) Interacting Members Apalutamide, CarBAMazepine, Encorafenib, Enzalutamide, Fosphenytoin, Lumacaftor and Ivacaftor, Mitotane, Phenytoin, RifAMPin

Discussion In healthy volunteer studies, rifampin (600 mg daily) decreased the AUC of immediate-release oral, extended-release oral, and IV tacrolimus by 68%, 55%, and 35%, respectively.[1,2]

Several case reports describe significant reductions in tacrolimus levels following the addition of rifampin requiring higher doses of tacrolimus to maintain therapeutic concentrations.[3,4,5,6,7] One case resulted in possible organ rejection 1 month after starting rifampin.[5] Other cases required the addition of itraconazole or ketoconazole or switching from rifampin to rifabutin to obtain therapeutic levels.[7,8,9]

A heart transplant recipient was found to have a 50% decrease in tacrolimus AUC to dose ratio and 45% increase in tacrolimus Cmin to dose ratio despite a 30% increase in tacrolimus dose following the addition of carbamazepine.[10]

A 57-year-old patient with history of seizure disorder treated with phenobarbital and carbamazepine underwent a living kidney transplant.[11] Following carbamazepine discontinuation, halving of the phenobarbital dose, and a 5-fold increase in tacrolimus dose, goal trough levels were achieved.

One case report describes 2 heart transplant recipients with tacrolimus serum concentration to dose ratios that were significantly lower during concomitant therapy with phenytoin compare with tacrolimus therapy alone.[12] Another describes a 54-year-old kidney transplant recipient who required tacrolimus 0.25 mg/kg/day while on phenytoin therapy to maintain therapeutic trough levels, but only required tacrolimus 0.16 mg/kg/day to achieve therapeutic concentrations after phenytoin discontinuation.[4] Other cases describe kidney transplant recipients who ha difficulty achieving therapeutic tacrolimus concentrations while on phenytoin therapy.[13,14] In addition to high tacrolimus dose requirements while on phenytoin therapy, one case report describes a kidney transplant recipient who experienced elevated phenytoin levels after his cyclosporine was changed to tacrolimus.[15]

Phenytoin has been used in numerous cases to enhance tacrolimus elimination in cases of tacrolimus toxicity.[16,17,18,19,20,21]

門診醫囑處方查詢-處方內容全覽

病歷號) 35.80 歲 49kg 168.5cm BSA： 1.514 0002 看診日 1080321 上午 感染科

序	類 別	處方日期	醫師姓名	作廢	作廢人員	作廢時間

【Subject Finding】
THIS PATIENT WAS DISCHARGED FOR OUR HOSPITAL
THE DISCHARGE DIAGNOSIS IS
- SEPSIS WITH METABOLIC ACIDOSIS WITH RESPIRATORY COMPENSATION, F
- SUSPECT HIV INFEECTION.
- LEUKOCYTOPENIA, ANEMIA,
- ELECTROLYTE IMBALANCE WITH HYPOOSMOLAR HYPONATREMIA/HYPERK
- HEPATITIS, CAUSE?
THIS PATIENT CONDITION WAS GROSSLY STATIONARY
108/03/21 陳昶華醫師記錄,患者 [未確知是否有 G6PD缺乏] 病史

ICD 疾病碼 顯示SOAP文字彙總

版本	序	疾病碼	主	次	疑	疾 病 名 稱 (英)
10	1	B20	Y	N	N	Human immunodeficiency virus [HIV] disea
10	2	A31.9	Y	N	N	Mycobacterial infection, unspecified
10	3	R25.1	N	N	N	Tremor, unspecified
10	4	R00.2	N	N	N	Palpitations
10	5	K27.9	N	N	N	Peptic ulcer, site unspecified, unspecified a

處方內容:

藥 品 項 目 (10) 非 藥 品 項 目 (17)

序	項	次	收費碼	藥品項目名稱	劑量	劑單	服法	天數	總銷量	銷
1	1	0	MTTUM	Abacavir/Lamivudine/Dolutegravir Tab. 600mg/300mg/50mg (...	1	tab	HS	7	7	tab
1	2	0	MTFAC	Folic acid 5mg/ tab (Folacin F.C) 5mg/ tab	1	tab	QD	7	7	tab
1	3	0	MTINH	【100mg/ tab】Isonicotinic acid hydrazide(Isoniazide) 100mg/ t...	3	tab	QD	7	21	tab
1	4	0	MTEMB	Ethambutol 400mg/ tab (E-BUTOL) 400mg/ tab	2	tab	QD	7	14	tab
1	5	0	MTRIF	Rifabutin Cap. 150mg/cap (Mycobutin Cap.) 150mg/cap	2	cap	QD	7	14	cap
1	6	0	MTETO	【錠】Esomeprazole TAB. 40mg/tab (NEXium TAB.) 40mg/tab	1	tab	QD	7	7	tab
1	7	0	MTVBSV	Pyridoxine(Vit.B6) Tab. 50mg/ tab (Weta B6 Tab.) 50mg/ tab	1	tab	QD	7	7	tab
1	8	0	MTBAK	Sulfamethoxazole 400mg & Trimethoprim 80mg 400mg/80mg/...	2	tab	HS	7	14	tab
1	9	0	MTCLYH	【錠】Clarithromycin TAB. 500mg/tab (Klaricid TAB.) 500mg/tab	1	tab	BID	7	14	tab
1	10	0	MTACMH	AcetaMINOPHEN TAB. 500mg/tab (Acetal TAB.) 500mg/tab	1	tab	ASO	7	10	tab

門診醫囑處方查詢-處方內容全覽

病歷號 (男) 35.80 歲 49kg 168.5cm BSA : 1.514 看診日 1080328 上午 員基感染科

序	類 別	處方日期	醫師姓名	作廢	作廢人員	作廢時間
總	醫師處方	1080328	張志演			
1	醫師處方	1080328	張志演	N		
2	電腦自動轉入	1080328	張志演	N		
3	事務處方	1080328	張志演	N		

【Subject Finding】
. HIV infection with Acquired Immunodeficiency syndrome stage with pulmo
pulmonary pneumocystis Jiroveci pneumonia and suspect fungal infection w
endotracheal tube obstruction ,re-intubation on 2019/01/07-01/26, s/p re-i
3. CMV infection
4. Multiple active GUS with SRH at antrum complicated with anemia s/p Bos
initial hemostasis on 2019/01/17
5. Pneumonia with respiratory failure post extubation (Sputum culture grew
6. Drug abuse (Synthetic cathinones with bath salts use)
7. Sacral pressure sore with secondary infection, improvement
8. Anemia, related to infection with poor intake

ICD 疾病碼 顯示SOAP文字彙總

版本	序	疾病碼	主	次	疑	疾 病 名 稱 (英)
10	1	B20	Y	N	N	Human immunodeficiency virus [HIV] di
10	2	A31.9	Y	N	N	Mycobacterial infection, unspecified
10	3	A15.0	Y	N	N	Tuberculosis of lung
10	4	R25.1	N	N	N	Tremor, unspecified
10	5	R00.2	N	N	N	Palpitations

處方內容:

藥 品 項 目 (9) 非 藥 品 項 目 (15)

序	項	次	收費碼	藥品項目名稱	劑量	劑單	服法	天數	總銷量
1	1	0	MTTUM	Abacavir/Lamivudine/Dolutegravir Tab. 600mg/300mg/50mg (...	1	tab	HS	7	7
1	2	0	MTFAC	Folic acid 5mg/ tab (Folacin F.C) 5mg/ tab	1	tab	QD	7	7
1	3	0	MTINH	【100mg/ tab】Isonicotinic acid hydrazide(Isoniazide) 100mg/ t...	3	tab	QD	7	21
1	4	0	MTEMB	Ethambutol 400mg/ tab (E-BUTOL) 400mg/ tab	2	tab	QOD	7	8
1	5	0	MTRIF	Rifabutin Cap. 150mg/cap (Mycobutin Cap.) 150mg/cap	2	cap	QD	7	14
1	6	0	MTETO	【錠】Esomeprazole TAB. 40mg/tab (NEXium TAB.) 40mg/tab	1	tab	QD	7	7
1	7	0	MTVBSV	Pyridoxine(Vit.B6) Tab. 50mg/ tab (Weta B6 Tab.) 50mg/ tab	1	tab	QD	7	7
1	8	0	MTBAK	Sulfamethoxazole 400mg & Trimethoprim 80mg 400mg/80mg/...	1	tab	HS	7	7
1	9	0	MTCLYH	【錠】Clarithromycin TAB. 500mg/tab (Klaricid TAB.) 500mg/tab	1	tab	BID	7	14

患者資料	處方(~3Mth)	DDI Info.	住院處方	藥歷軸	更早(~1Y)		列印: 英文 中文

年齡	69.4歲	Ht/Wt	158cm/51kg
---		行動	---

Bilirubin-T: 0.4(25/03/21)	Creatinine: 0.54(25/03/21)	CrCl: 92.74mL/min	Estimated GFR: 111.94(25/03/21)
GOT (AST): 22(25/03/21)	GPT (ALT): 10(25/03/21)	Bilirubin-T: 0.4(24/12/12)	GPT (ALT): 10(24/12/12)
GOT (AST): 17(24/12/12)	Estimated GFR: 125.59(24/12/12)	Creatinine: 0.49(24/12/12)	CrCl: 102.21mL/min
Bilirubin-T: 0.4(24/11/11)	GPT (ALT): 7(24/11/11)	GOT (AST): 20(24/11/11)	Estimated GFR: 135.09(24/11/11)
Creatinine: 0.46(24/11/11)	CrCl: 108.87mL/min	Creatinine: 0.46(24/08/22)	CrCl: 108.87mL/min
GPT (ALT): 11(24/08/22)	Estimated GFR: 135.09(24/08/22)	Bilirubin-D: 0.09(24/08/22)	

感染科-張志演(mvpn:672024) 第211診

2025-03-03 17:39(一般)

- A15.0 Tuberculosis of lung
- B00.89 Other herpesviral infection
- B20 Human immunodeficiency virus [HIV] disease
- B59 Pneumocystosis
- F01.50 Vascular dementia, unspecified severity, without behavioral disturbance, psychotic disturbance, mood disturbance, and anxiety
- F02.80 Dementia in other diseases classified elsewhere, unspecified severity, without behavioral disturbance, psychotic disturbance, mood disturbance, and anxiety
- G30.9 Alzheimer's disease, unspecified
- J18.9 Pneumonia, unspecified organism
- K75.9 Inflammatory liver disease, unspecified

1. 5g(【外用】Rinderon VA CREAM)	1	BID-TOPI	28	3**tube**
2. 20g(美康乳膏)	1	BID-TOPI	28	3**tube**
3.Pyridoxine(Vit.B6) Tab. 50mg/ tab(Weta B6 Tab.)	1tab	QD-PO	28	28**tab**
~~**4.**Abacavir/Lamivudine/Dolutegravir Tab. 600mg/300mg/50mg(Triumeq Tab.)~~	1tab	QD-PO	28	28**tab**
~~**5.**RIF 300mg/INH 150mg 300mg/cap(RINA (二合一))~~	2cap	QD-PO	28	56**cap**
6.Prednisolone TAB. 5mg/ tab(Compesolon TAB.)	0.5tab	QD-PO	28	14**tab**
7.Mosapride citrate 5mg/ tab(Mosad Tab)	1tab	TID-PO	28	84**tab**
8.Levocetirizine 5mg/tab(Xyzal)	1tab	QD-PO	28	28**tab**
9.Folic acid 5mg/ tab(Folacin F.C)	1tab	QD-PO	28	28**tab**
10. Ethambutol 400mg/ tab(E-BUTOL)	2tab	QD-PO	28	56**tab**
~~**11.**Dolutegravir Tab.(急採) 50mg/tab(Tivicay Tab.(急採))~~	1tab	HS-PO	28	28**tab**
12.【錠】Clarithromycin TAB. 500mg/tab(Klaricid TAB.)	1tab	BID-PO	28	56**tab**

門診醫囑處方查詢-處方內容全覽

看診日 1140522 上午 員基感染科 張志演

序	類 別	處方日期	醫師姓名	作廢	作廢人員	作廢時間
總	醫師處方	1140522	張志演			
1	醫師處方	1140522	張志演	N		
2	醫師處方	1140522	張志演	N		

【Subject Finding】
1.Large amount pericardial effusion s/p pericardiocentesis on 2025/04/15, removed pericardial effusion drainage on 2025/04/17, etiology: related to uremia or autoimmune disease ?#
2.Decompensated heart failure, etiology: CAD, valvular heart disease#
3.Conscious drowsiness casue not determine, should be ruled out seizure attack#
4.AV shunt dysfunction s/p PTA for left cephalic vein
5.Pneumonia
6.ESRD with hyperkalemia, renal anemia#

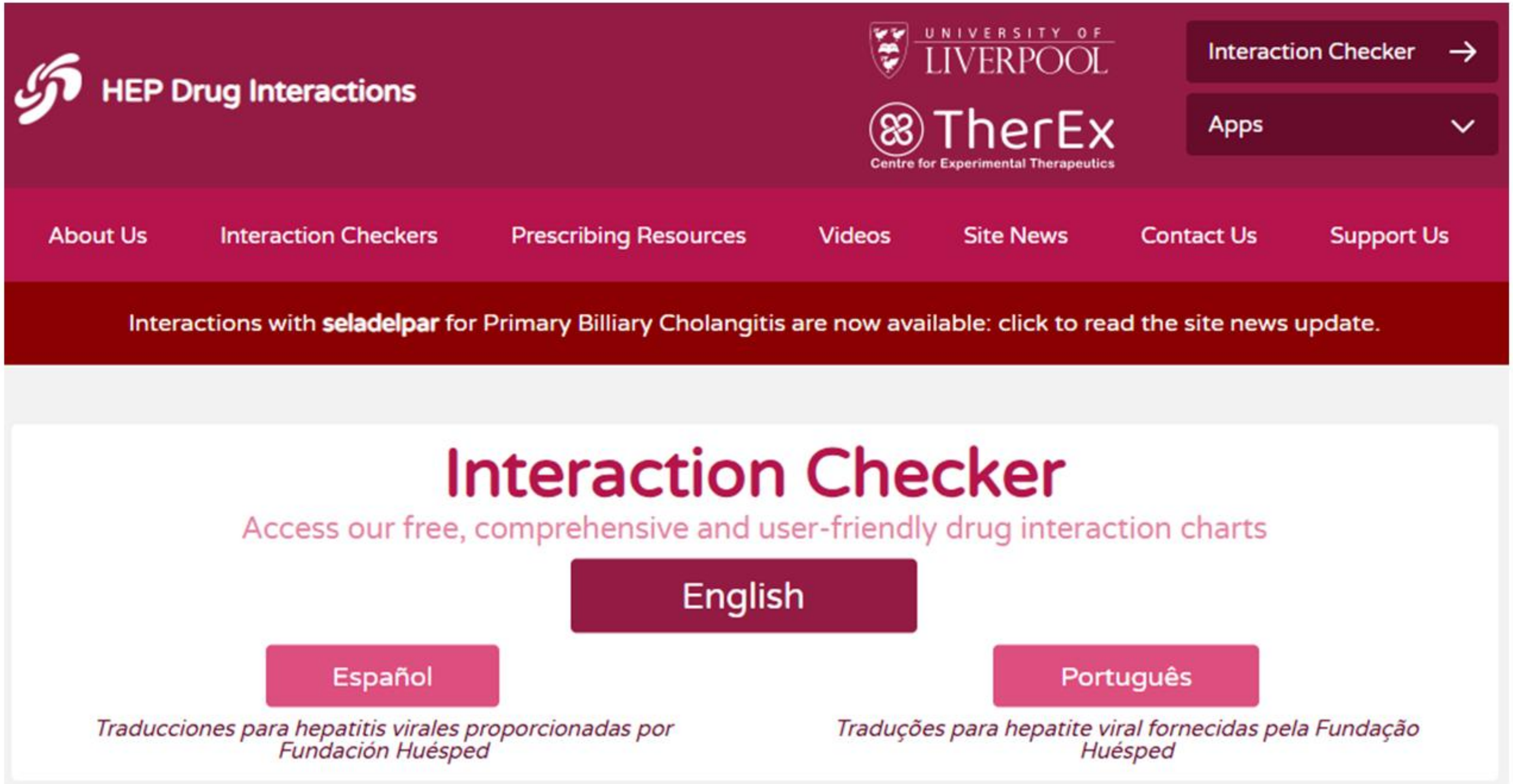
ICD 疾病碼 | 顯示SOAP文字彙總

版本	序	疾病碼	主	次	疑	疾 病 名 稱 (英)	疾 病 名 稱 (中
10	1	N18.6	Y	N	N	End stage renal disease	末期腎疾病
10	2	A18.84	N	Y	Y	Tuberculosis of heart	心臟結核
10	3	E11.21	N	Y	N	Type 2 diabetes mellitus with diabetic nephropathy	第二型糖尿病

處方內容:

藥 品 項 目 (4) | 非 藥 品 項 目 (11)

序	項	次	收費碼	藥品項目名稱	劑量	劑單	服法	天數	總銷量	銷單	剩餘量	途徑
1	1	0	MTRFIH	RIF 300mg/INH 150mg 300mg/cap (RINA (二合一)) 300mg/cap	2	cap	QD	14	28	cap	0	PO
1	2	0	MTPRMH	Pyrazinamide 500mg/tab (Pyrazinamide) 500mg/tab	2	tab	W135	14	12	tab	0	PO
1	3	0	MTEMB	Ethambutol 400mg/ tab (E-BUTOL) 400mg/ tab	2	tab	W135	14	12	tab	0	PO
1	4	0	MTVBSV	Pyridoxine(Vit.B6) Tab. 50mg/ tab (Weta B6 Tab.) 50mg/ tab	1	tab	QD	14	14	tab	0	PO


HEP Drug Interactions
UNIVERSITY OF LIVERPOOL
TherEx
Centre for Experimental Therapeutics
Interaction Checker
Apps
About Us
Interaction Checkers
Prescribing Resources
Videos
Site News
Contact Us
Support Us
Interactions with seladelpar for Primary Billiary Cholangitis are now available: click to read the site news update.
Interaction Checker
Access our free, comprehensive and user-friendly drug interaction charts
English
Español
Traducciones para hepatitis virales proporcionadas por Fundación Huésped
Português
Traduções para hepatite viral fornecidas pela Fundação Huésped

資料庫/電子期刊/電子書 ? › 查詢紀錄

彰化基督教醫院圖書館
電子資源管理系統

全部 | 查詢

› 進階查詢 › 資源瀏覽

Sanford Guide 熱病資料庫

新訂

資源網址: https://webedition.sanfordguide.com/en

語文別: 西文

使用範圍: 總院院內直接使用, 院外需認證

公告資訊: Sanford Guide熱病為傳染病/抗生素用藥主題電子書，該資料庫亦有提供APP使用。(**使用手冊**請點此資料庫名稱進入詳目頁，參考相關附檔)

摘要: Sanford Guide熱病為傳染病/抗生素用藥主題電子書，該資料庫亦有提供APP使用。

是否使用proxy: 是

資料類型: 圖書/電子書

資源簡介

Sanford Guide熱病為傳染病/抗生素用藥主題電子書，包括有各系統器官感染疾病的常見病原體、傳播途徑、診斷要點、首選和備選治療方案、藥物不良反應和應用注意事項，以及預防用藥等，並輔以有關文獻來體現實證醫學的權威性；涉及的病原體有細菌、真菌、寄生蟲和病毒等。

內容提供三大主題，包含**Antimicrobial Therapy**、**HIV/AIDS Therapy** 及 **Hepatitis Therapy**，全新排版，依主題瀏覽，提供互動計算功能，每月定期線上更新，依據最新證據提供最新醫藥發展。支援多個關鍵字搜索、Spectra of Activity、計算機（BMI, Ccr, BSA, PSI等）、表格、個人化書簽等。該資料庫有提供APP使用。

相關檔案

Sanford Guide Collection使用手冊_手機版 Sanford Guide Collection使用手冊_網頁版

相關連結

影音介紹

熱病 SANFORD GUIDE ANTIMICROBIAL

Syndromes | Pathogens | Anti-infectives | Prevention | Tables & Tools

QUICK LINKS

- Preview the Guide
- Spectra of Activity
- Regional Guidelines
- View Latest ID Updates
- Duration of Therapy
- Measles
- Pharmacology
- Drug Usage & Dosing
- Vaccines

ID Update

Subscribe to updates

MAY 2025

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Article of the Month (Editors' Choice)

Cellular HIV DNA Resistance Testing

By Michael S. Saag, MD

- In a recent Viewpoint article in *Clinical Infectious Diseases* (Clin Infect Dis 2025;Apr 3:ciaf161 [online ahead of print]; click here for PDF), Wensing and colleagues review the **benefits and limitations of using cellular HIV DNA resistance testing** in clinical practice.* The ability using PCR to amplify and sequence HIV DNA from integrated cellular DNA has been available commercially for over a decade, and has been more frequently used since the introduction of long-acting injectable antiretroviral (LA-ARV) drugs into clinical practice. The formal indications for use of LA-ARV require

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- HIV Risk Assessment
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- Meningococcal B, Vaccines
- Chikungunya, Vaccine
- RSV, Vaccine
- Obesity Dosing Adjustments

UpToDate-Drug interaction

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< Back **Drug Interactions** Print

Item(s)

Enter Item Name Add

✕ Doravirine, Lamivudine, and Tenofovir Disoproxil Fumarate

✕ RifAMPin

Clear Analyze

Display complete list of interactions for an individual item by clicking item name.

NOTE: This tool does not address chemical compatibility related to I.V. drug preparation or administration.

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Title CYP3A4 Inducers (Strong) / Doravirine

Risk Rating X: Avoid combination

Summary CYP3A4 Inducers (Strong) may decrease serum concentration of Doravirine. **Severity** Major **Reliability Rating** Intermediate

Patient Management Do not use doravirine with strong CYP3A4 inducers. This combination is listed as a contraindication in the doravirine prescribing information. Discontinue strong CYP3A4 inducers at least 4 weeks before initiation of doravirine.

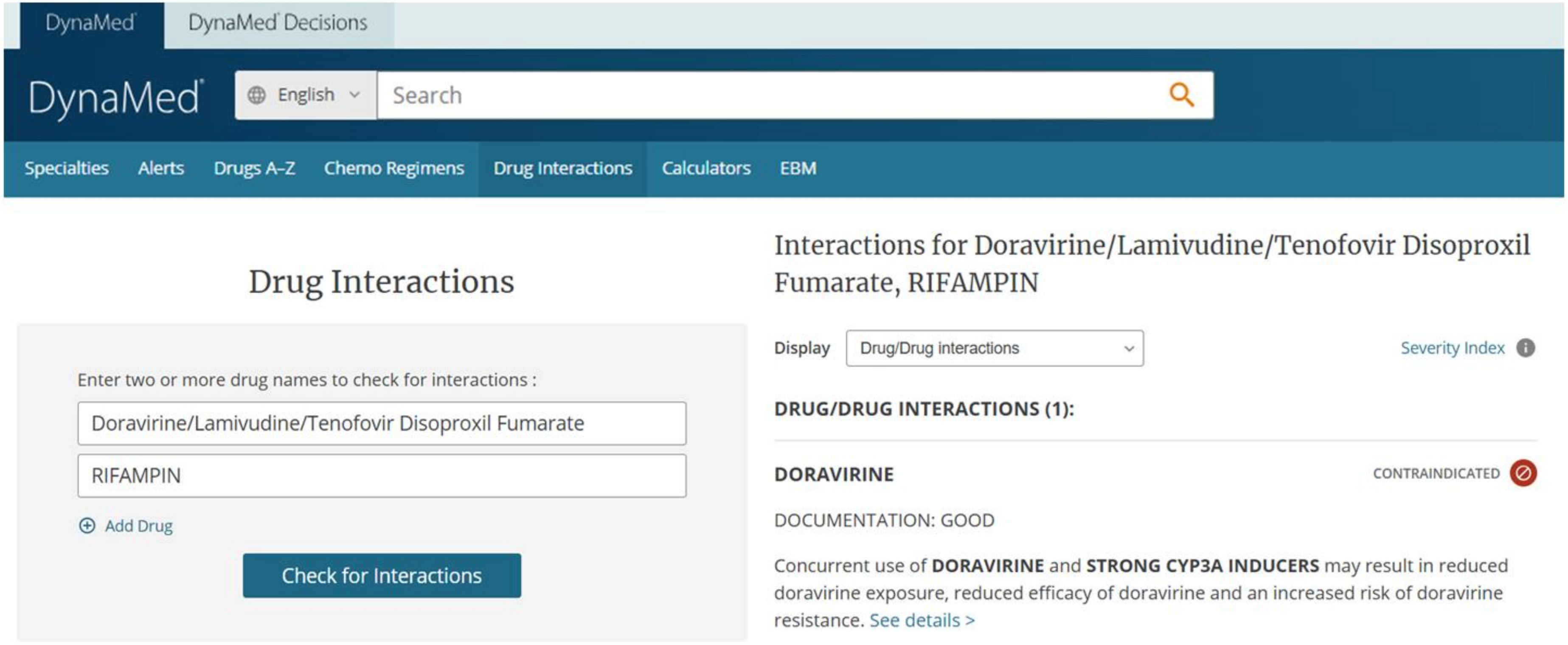
CYP3A4 Inducers (Strong) Interacting Members Apalutamide, CarBAMazepine, Encorafenib, Enzalutamide, Fosphenytoin, Lumacaftor and Ivacaftor, Mitotane, Phenytoin, RifAMPin

Discussion In a pharmacokinetic study of 10 healthy volunteers, the strong CYP3A4 inducer rifampin (600 mg daily for 15 days) decreased the doravirine (100 mg single dose given on day 14) AUC, maximum serum concentration, and minimum serum concentration 88%, 57%, and 97%, respectively.[1]

Doravirine prescribing information states that because strong CYP3A4 inducers significantly decrease doravirine serum concentrations and may lead to decreased effectiveness, concomitant use of doravirine and strong CYP3A4 inducers is contraindicated.[2,3]

The mechanism of this interaction is likely rifampin-mediated induction of CYP3A4, an enzyme responsible for doravirine metabolism.

DynaMed-Drug interaction



Take Home Message

- 衛生福利部疾病管制署結核病診治指引
- 極端肝腎功能注意藥品劑量及調整

Pyrazinamide、Ethambutol須調整劑量

Pyrazinamide、Isoniazid、Rifampin肝毒性較強

- 肥胖病人應注意劑量調整:記得輸入身高及體重!!

Rifampin: TBW (Max:600mg)

Ethambutol、Isoniazid、Pyrazinamide: IBW

- B肝、C肝、愛滋、抗Covid-19病毒藥物交互作用
- 找一個配合良好的團隊(個管師、藥師、護理師、專科護理師)

結核病診治指引

Taiwan Guidelines for TB Diagnosis & Treatment

第七版

衛生福利部疾病管制署　編

主　　編　江振源


INFECTIOUS DISEASE CLINICAL PHARMACIST
COLLABORATION WITH TEAM

AI and Machine in TB medication

- X光診斷
- 抗藥性的預測
- GWAS (Taiwan bioBank)
- 新型藥物的開發
- 副作用預測

NEWS | 27 March 2024

An app that hears could be first point for TB screening

Algorithm is better than the human ear at distinguishing between similar cough frequencies

TB Medication in Critical illness patient

- Body weight change
- Unpredictable ADR and subtherapeutic drug
- Unable to swallow (Decrease efficacy)
- Hire a pharmacist!!!!!